No Language Left Behind: Scaling Human-Centered Machine Translation

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Abstract

Driven by the goal of eradicating language barriers on a global scale, machine translation has solidified itself as a key focus of artificial intelligence research today. However, such efforts have coalesced around a small subset of languages, leaving behind the vast majority of mostly low-resource languages. What does it take to break the 200 language barrier while ensuring safe, high quality results, all while keeping ethical considerations in mind? In *No Language Left Behind*, we took on this challenge by first contextualizing the need for low-resource language translation support through exploratory interviews with native speakers. Then, we created datasets and models aimed at narrowing the performance gap between low and high-resource languages. More specifically, we developed a conditional compute model based on Sparsely Gated Mixture of Experts that is trained on data obtained with novel and effective data mining techniques tailored for low-resource languages. We propose multiple architectural and training improvements to counteract overfitting while training on thousands of tasks. Critically, we evaluated the performance of over 40,000 different translation directions using a human-translated benchmark, Flores-200, and combined human evaluation with a novel toxicity benchmark covering all languages in Flores-200 to assess translation safety. Our model achieves an improvement of 44% BLEU relative to the previous state-of-the-art, laying important groundwork towards realizing a universal translation system. Finally, we open source all contributions described in this work, accessible at `https://github.com/facebookresearch/fairseq/tree/nllb`.

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†. Research and engineering leadership, equal contribution, alphabetical order

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Contents

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2.1 Exploratory Interview Study Research Design 8
2.2 No Language Left Behind: Guiding Principles 11

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4.1 FLORES-200 18
4.2 NLLB Seed Dataset 23
4.3 NLLB Multi-Domain Dataset 24
4.4 Conclusion 25

U 8«§nh9§_H9ee´ G}09§_jX ¦}9j e9§_nj ¦}9_j_jX K9§9 Tn} [«jL}0L nT d9j] X«9X0 ©U
5.1 Language Identification 26
5.2 Gathering and Cleaning Monolingual Data at Scale 34
5.3 Mining Bitexts for Low-Resource Languages 37
5.4 Conclusion 46

£ gnL0e_jX V¢
6.1 Preliminaries 48
6.2 Conditional Compute for Massively Multilingual Machine Translation . . . 50
6.3 Self-Supervision Strategies on Large-scale Monolingual Corpora 59
6.4 Data Augmentation 64
6.5 Bootstrapping models with NLLB-Seed 67
6.6 Human Evaluation 70
6.7 Conclusion 71

¢ N®9e«9§_nj ¢o
7.1 Automatic Evaluation 72
7.2 Human Evaluation 73
7.3 Toxicity 79
7.4 Conclusion 89

P ?}_jX_jX_§ 8ee ¦nX0§\0} k·
8.1 Preparing the Data 91
8.2 Preparing the Model 97
8.3 Results on Flores-200 102
8.4 Out-of-domain Generalization: Performance on non-Flores-200 Domains . 108
8.5 Analysis of NLLB-200 114
8.6 Making Large Models More Accessible through Distillation 116

3Xd 1z2+iBp2Hv AM+Hm/BM; G M;m ;2b rBi? JmHiBTH2ya+`BTib M
3X3 1MpB`QMK2Mi H AKT +i Q7 LGG" X X X X X X X RkX X X X X X

N LQ G M;m ;2 G27i "2?BM/, aQ+B H AKT +i *QM+Hm/BM; h?Qm;?ib
NXR 1tT M/BM; AM7Q`K iBQM ++2bb X X X X X X XRXkNX X X X X X
NXk h?2 C Mmb@7 +2/ L im`2 Q7 .B;Bi H S `iB+BT iBRQkM X X X X X
NXj h?2 6mim`2 Q7 LGG", *QHH2+iBp2 _2bTQMbB#BHJBiv X X X X

Ry *QMi`B#miBQMb RjR

RR +FMQrH2/;2K2Mib Rjj

G M;m ;2b Red

" 1p Hm iBQM Re3

* . i Rdy

. JQ/2HBM; Rd9

1 "`BM;BM; Bi HH hQ;2i?2` Rd8

6 JQ/2H * `/ @ LGG"@kyy R39

: . i * `/ 7Q` LGG"@a22/ . i R3d

> . i * `/ 7Q` LGG" JmHiB@.QK BM . i R3N

A . i * `/ 7Q` JBM2/ "Bi2ti J2i / i RNR

AM C +F o M+2 URNddVh?2 b1+vB2@ }QM Q?2 H-pB2ibQT``HQ/i ;QMBbi- *m;2H-
2M+QmMi2`b rBx `/r?Q +QKT2Hb ?BK BMiQ i bFX hQ bbBbi?BK-i?2
K ;B+ H /2AM+Q2`/2` iQ 7 +BHBi i2 vQm` bT22+?- A 2M/Qr vQm rBi? i?B
`2H i2b HH TQbbB#H2 pQ+ #H2b iQ 2p2`v +QM+2Bp #H2 bvbi2K Q7 K2

6 bi@7Q`r `/? H7 +2Mim`v H i2`-r2 MQr FMQr i? i*m;2Höb K ;B
J +?BM2 h` MbH iBQMX *QM+2Bp2/ b +QKTmi iBQM H bvbi2Kb i? ii
H M;m ;2 iQ MQi?2`-K +?BM2 i` MbH iBQM ? b #22M `QmM/ bBM+2
KB;` iBQM 7`QK bi iBbiB+ H U"`QrM 2i HX-RNNjc EQ2?M-kyyNc GQT2
? b Tmb?2/ i?2 i2+?MQHQ;v iQ M2r 7`QMiB2`b U" ?/ M m 2i HX-ky
E H+?#`2MM2` M/ "HmMbQK-kyRjc qm 2i HX-kyReVX h?Bb b?B7i
i` MbH iBQM [m HBiv i#`2 FM2+F bT22/-#mi Bi? b HbQ 7m`i?2`2/ i
i` MbH iBQM BMiQ M2r TTHB+ iBQMbX hQ/ v-K +?BM2 i` MbH iBQM
i?2 rQ`H/ +QKKmMB+ i2-rQ`F-i` p2H-H2 `M- ++2bb BM7Q`K iBQM
_Q/`B;m2x-kykkc EQ2?M M/:2`K MM-kyR9c G22-kykyVX

q?BH2 K +?BM2 i` MbH iBQM +QMiBMm2b iQ ;`Qr-i?2 7`mBib Bi
i`B#mi2/U6 M 2i HX-kykyVX AM 7 +i-i?2 p biK DQ`Biv Q7 BKT`Qp
i` MbH iBQM BM i?2 H bi/2+ /2b ? p2 #22M 7Q` ?B;?@`2bQm`+2 H M;m
H `;2 [m MiBiB2b Q7 i` BMBM;/i p BH #H2/B;Bi HHvX 6Q` BMbi M
BM 1M;HBb?-6`2M+?-:2`K M Q` _mbbB M H M;m ;2b r?B+? ? p2 HQM
BMp2biK2Mib M//i p BH #BHBiv bi M/iQ ; BM bm#bi MiB HHv KQ`2
K +?BM2 i` MbH iBQM i? M i?Qb2 r?Q bT2 F* i H M- bb K2b2-GB;m`

J Mv H M;m ;2b Q7 i?Bb H ii2`;`QmT ii` +i H2bb ii2MiBQM M/`2
KQbi H M;m ;2b bTQF2M ;HQ@r@H_2bQmV;m22;2b UCQb?B 2i HX-kykyVX
Q7 i?2b2 H M;m ;2b 2b+ T2 `2b2 `+?2`bö ; x2 7Q` +QM~m2M+2 Q7 `2
+QMDm`2/mT #v T bi BMp2biK2Mib UQ` H +F i?2`2Q7V-`2b2 `+? MQ`K
M/q2bi2`M@+2Mi`BbK iQ M K2 72rX qBi?Qmi M 2zQ`i iQ +Qm`b2
BMi2`M2i +QmH/ +QMiBMm2 iQ #2 BM ++2bbB#H2 iQ bT2 F2`b Q7 i?2b2
i? ir?BH2 QMHvk8XN T2`+2Mi Q7 BMi2`M2i mb2`b bT2 F 1M;HBb?-ejXd
1M;HBb? Ui?2 M2ti QM i?2 HBbi Bb _mbbB M ieX3 T2`+2Mic _B+?i2`-ky
H M;m ;2 +QKKm?2BSQ2Hv;HQiAWm2+FM22K M-kyy3V- M BMbi`mK2Mi H K
i? i+QmH/ T`QT2H 2/m+ iBQM ++2bb M/bQ+B H KQ#BHBiv-`2K BM
r2# ? b HQM; T`BQ`BiBx2/ +QMi2Mi i BHQ`2/iQ ?B;?@`2bQm`+2 H M;m

1tT M/BM; K +?BM2 i` MbH iBQM iQ KQ`2 HQr@`2bQm`+2 H M;m ;2
i2+?MB+ H +? HH2M;2b U> //Qr 2i HX-kykkVX *QKT `2/iQ i?2B` ?B;
i` BMBM;/i 7Q` HQr@`2bQm`+2 H M;m ;2b `2 2tT2MbBp2 M/HQ;B
+m`2 UEmr MiQ 2i HX-kykRc L2FQiQ 2i HX-kykyc P`B72 2i HX-k
i` BMBM;/i -bi M/ `/i2+?MB[m2b K v MQi bi M/i?2 i2bi Q7 2K2`;BM
?m`/H2b ? p2 #2+QK2 2p2` KQ`2 T`QMQmM+2/ b i?2 TQTmH `Biv Q7
bm+? b H `;2@b+ H2 T`2@i` BMBM; M/KQ/2H b+ HBM;? p2 #2+QK2
G KTH2-kyRNc *QMM2 m 2i HX-kykyc E2MiQM M/hQmi MQp-kyRNc

hQ Qp2`+QK2 i?2b2 # ``B2`b-Km+? 2tBbiBM; rQ`F QM HQr@`2bQm`+
QM H2p2`KmBMiBMbWm2Hb-Q` KQ/2Hb + T #H2 Q7 ? M/HBM; KmHiBTH2 H
KQ/2Hb ? p2 i?2 /p Mi ;2 Q7 +`QbbHBM;m H i` Mb72` UL;mv2M M/*?

6B;m`2>kQ,r i?2 SB2+2b 6Bi hQ;2i?2`- "B`/öq2@/12vT2Be+B1?2 i2+?MB+ H
+QKTQM2Mib Q7 LQ G M;m ;2 G27i"2?BM/ M/ ?Qr i?2v }i iQ;2i?2`X q2
BMi2` +iBQM #2ir22M / i -?Qr / i Bb miBHBx2/ BM i?2 KQ/2Hb r2 /2p2
KQ/2Hb `2 2p Hm i2/X . i b2ib b?QrM BM #Hm2 `2 MQp2H / i b2ib +`2
G27i"2?BM/X

?mK M 2p Hm iBQMbX q2 T`QTQb2/ M/ +QM/m+i2/ ?mK M 2p Hm iBQMb
KQ/2Hb +Qp2`- BM //BiBQM iQ i?2 +QKKQM miQK iB+ K2i`B+b- iQ ; Bb
i?2 BKT +i Q7 i?2 i` MbH iBQMX 6BM HHv- #2vQM/ +`2 iBM; i?2b2 KQ/
+`2 iBQM T`Q+2bb- M HvxBM; i?2 `BbFb M/ #2M2}ib Q7 Qm` `2b2 `+?
q2 QT2M bQm`+2 HH i?2 #2M+?K `Fb-/ i -b+`BTib- M/ KQ/2Hb /2b+
bmTTQ`i 7m`i?2R A2Mb2//B+iBXQM- r2 7Q+mb QM i?2 T` +iB+ H TTHB+ #BH
HQr@`2bQm`+2 bT2 FBM; +QKKmMBiB2bX q2 /2THQv Qm`i2+?MB[m2b i
iQ qBFBT2/B 2/BiQ`b- 2M #HBM; i?2K iQ +`2 i2 M2r `iB+H2b KQ`2 2
i? i `2 MQi bmTTQ`i2/ #v Qi?2` i` MbH iBQM bvbi2KbX

h?2 `2bi Q7 i?2 `iB+H2 Bb bi`m+im`2/ b 7QHHQrb- rBi? 6B;m`2 k b
/2b+`B#2b i?2 QT2M +? HH2M;2b BM HQr@`2bQm`+2 i` MbH iBQM M
Q7 i` MbH iBQM bvbi2KbX a2+iBQM j T`2b2Mib i?2 H M;m ;2b r2 7Q+
i i?Bb b2i Q7 H M;m ;2bX a2+iBQM 9 bmKK `BxH2Q`2b2@+Mv2y iBQM T`Q
LGG"@a2Y2LGG"@J-.Qm` i` MbH iBQM b22//i b2ib-rBi?[m HBiv M Hv
Qp2`pB2rb i?2 +`2 iBQM Q7 KQMQHBM;m H M/ KBM2/#BHBM;m H / i -
Q7 KQ/2Hb 7Q`?mM/`2/b Q7 H M;m ;2bX a2+iBQM e/2i BHb p `BQmb KQ
iQ BKT`Qp2 i?2 T2`7Q`K M+2 Q7 HQr@`2bQm`+2 H M;m ;2bX a2+iBQM
?mK M 2p Hm iBQM Q7 Qm` i` MbH iBQMb- BM+Hm/BM; i?2 /2i2+iBQ
i` MbH iBQMbX q2 BMi2;` i2 i?2 7Q`2K2MiBQM2//i b2ib M/ i2+?MB

RX HH `2 p BH?#HT2b?f2;Bi?m#X+QKf7 +2#QQF`2b2 `+?f7 B`b2[fi`22fMHH#

KQ/2H i? i +m``2MiHv bmTTQ`ib kyk H M;m ;2b- M/ M Hvx2 Bib [m H
a2+iBQM 3X q2 +QM+Hm/2 BM a2+iBQM N- r?2`2 r2 `2~2+i QM i?2 bQ+
M/ H v Qmi 7mim`2 TQbbB#BHBiB2b M/ +? HH2M;2bX Ai Bb Qm` ?QT
;mB/2 7mim`2 `2b2 `+?2`b r?Q- HBF2 mb- `2 2 ;2` iQ b22 *m;2Höb K
i` MbH iBQM +Qp2`BM; HH H M;m ;2b i` Mb7Q`K 7`QK +QM+2Tim H
hQ K F2 Qm` rQ`F p BH #H2 iQ i?2 +QKKmMBiv- r2 QT2M bQm`+2 i?

Ç >mK M@h` MbH i2/ . i b2ib

6HQ`2b@kyy, kyyHm iBQM / i b2i BM ky9 H M;m ;2b

LGG"@a2,2a/22/ i` BMBM; / i BM jN H M;m ;2b

LGG"@J.a22/ / i BM /Bz2`2Mi /QK BMb BM e H M;m ;2b iQ bb2bb

hQtB+Biv@kyy, rQ`/HBbib iQ /2i2+i iQtB+Biv BM kyy H M;m ;2b

Ç hQQHb iQ *`2 i2 G `;2 a+ H2 "Bi2ti . i b2ib

G M;m ;2 A/2MiB}+ iBQM 7Q` KQ`2 i? M kyy H M;m ;2b

G a1_j, b2Mi2M+2 2M+Q/2`b 7Q` B/2MiB7vBM; HB;M2/ #Bi2ti 7

biQT,2b/ i KBMBM; HB#` `v i? i + M #2 mb2/ iQ T`Q+2bb M/ +H2
/ i - i?2M +`2 i2 HB;M2/ #Bi2ti

h` BMBM; / i `2+`2 iBQM, a+`BTib i? i `2+`2 i2 Qm` i` BMBM; /

Ç h` MbH iBQM JQ/2Hb +Qp2`BM; kyk H M;m ;2b

LGG"@kyy89X8" aT `b2Hv : i2/ JBtim`2@Q7@1tT2`ib KQ/2H

jXj" M/ RXj" .2Mb2 h` Mb7Q`K2` KQ/2Hb

RXj" M/ eyyJ .2Mb2 i` Mb7Q`K2` KQ/2Hb"@kyy

h` BMBM; M/ ;2M2` iBQM b+`BTib iQ `2T`Q/m+2 Qm` KQ/2Hb

kX >mK M@*2Mi2`2/ GQr@_2bQm`+2 G M;m ;2 h` MbH iBQM

hQ bBim i2 Qm` ;Q H Q7 T`QpB/BM; ?B;?@[m HBiv i` MbH iBQM 7Q` ?m
2tTHQ`2 i?2 BKTQ`i M+2 Q7 i?Bb `2b2 `+? iQ i?Qb2 r?Q K ii2` i?2 KQ
H M;m ;2 +QKKmMBiB2bX HmaTBb2 #Bi2 BK; M M/ >2M/`v- kyRNc
o M .2` >Qp2M M/ J M/2`b@>mBib- kykyV- r2 ii`B#mi2 +QKKmMBiv@H2
b i?2 +Q`M2`biQM2 Q7 Qm` `2b2 `+?X /QTiBM; i?Bb 7` K2rQ`F T`QT2H
M/ T`BQ`BiBx2 ?Qr i?2v BMi2` +i rBi? i2+?MQHQ;v- rBi? /B`2+i 2KT? b
+QMbB/2` iBQMb UJmF?BD 2i HX- kykRVX hQ mM/2`bi M/ ?Qr HQr@
T2`+2Bp2 K +?BM2 i` MbH iBQM- r2 +QM/m+i2/ M BMi2`pB2r bim/v rBi?
bT2 F2`bX b bi F2?QH/2`b HBF2Hv iQ #2 BKT +i2/ #v LQ G M;m ;2 G27
+QMi`B#miBQMb ?2HT2/ mb 2MpBbBQM i?2 T`QKBb2b K Mv #2HB2p2 K
iQ i?2B` +QKKmMBiB2bX SmM+im iBM; i?2B` + `27mH QTiBKBbK r2
r vb iQ K tBKBx2 bQ+B H ; BMb r?BH2 KBMBKBxBM; `BbFbX JQ`2Qp2`-
BHHmbi` iBp2 TB+im`2b Q7 i?2 +mHim` H M/ TQHBiB+ H 2MpB`QMK
r vb BM r?B+? H M;m ;2 M/ bQ+B H 2tT2`B2M+2b BMi2`irBM2- M/ ?Qr
b? F2 mT i?2 +mHim` H bi imb [mQX

kXR 1tTHQ` iQ`v AMi2`pB2r aim/v _2b2 `+? .2bB;M

q2 /2bB;M2/ b2KB@bi`m+im`2/ BMi2`pB2r T`QiQ+QH BK2/ i2tTHQ`BM
Q7 HQr@`2bQm`+2 H M;m ;2 bT2 F2`b pBb@¨@pBb K +?BM2 i` MbH
H M;m ;2b +QmH/ #2 /22K2/ HQr@`2bQm`+2 7Q` p `B2iv Q7 `2 bQM
`2b2 `+?2/- /B;BiBx2/- Q` i m;?i U*B2`B 2i HX- kyRec J ;m2`2bb2 2i H
Q7 i?2 bim/v- r2 /2}M2 HQr@`2bQm`+2 b H M;m ;2b r?B+? ? / H2bb i? I
Tm#HB+Hv p BH #H2 2t KTH2 i` MbH iBQMb i i?2 iBK2 Q7 i?2 bim/vX
#`Q / `` v Q7 iiBim/2b M/ mM/2`bi M/BM;b- BM+Hm/BM; i?2 mb ;2
`2bQm`+2 H M;m ;2b- T2`+2Bp2/ p Hm2 Q7 i` MbH iBQM i2+?MQHQ;v-
Qm;?i iQ #2 /2p2HQT2/X

Pp2` HH- Qm` `2+`mBiK2Mi 2zQ`i H2/ mb iQ 99 M iBp2 bT2 F2`b Q7
7`QK /Bp2`b2 # +F;`QmM/b- rBi? ;2b ` M;BM; 7`QK kj iQ 83X *Qp2`BM;
i?2 /Bbi`B#miBQM Bb b 7QHHQrb, 8 H M;m ;2b `2 bTQF2M T`2/QKBM
aQmi? K2`B+ - 9 BM 1m`QT2- Rk BM 7`B+ - M/ d BM bB X Hi?Qm;? C
BM i2`Kb Q7 ` +2- 2/m+ iBQM- M/ HQ+ iBQM- i?2 K DQ`Biv Q7 Qm` T
HBpBM; BM i?2 lXaX M/ 1m`QT2- M/ n#Q7mM Bi2BM/BQ7 i?2 K+U? rQ`F2`bX
HH BMi2`pB2rb r2`2 +QM/m+i2/ `2KQi2Hv pB pB/2Q +QM72`2M+BM;
BMi2`pB2rb H bi2/ RX8 ?Qm`bX hrQ@i?B`/ Q7 i?2 BMi2`pB2rb r2`2 `2+C
mM`2+Q`/2/ BMi2`pB2rb- irQ `2b2 `+?2`b iQQF 2ti2MbBp2 MQi2b i?`
BMi2`pB2rb iQ;2i?2`- `2bTQMb2b r2`2 i?2M bvbi2K iB+ HHv +Q/2/ iQ
B/2 b iQ 2K2`;2X

q2 +FMQrH2/;2 i? i b KTHBM; HQr@`2bQm`+2 H M;m ;2 bT2 F2`b 7
+QK2b rBi? Bib HBKBi iBQMbX 6Q` QM2- b BKKB;` Mib- i?2B` T2`bT2+i
+ Tim`2 i?2 b2MiBK2Mib Q7 i?2B` +QKKmMBiB2b # +F ?QK2X h? i b
`;m2/ i? i BM i2+?MQHQ;B+ HHv mM/2`/2p2HQT2/ M iBQMb- r?2`2 K
+QKKmMBiB2b `2bB/2- T2QTH2 i2M/ iQ pB2r i2+?MQHQ;v KQ`2 QTiBK
i? M i?Qb2 r?Q HBp2 BM TH +2b rBi? ?B;?2` H2p2Hb Q7 i2+?MQHQ;
2i HX- kykkc EQxv`2p 2i HX- kykRc a K# bBp M- kykRc a K# bBp M 2i
2tTQb2/ iQ +`BiB+ H i2+?MQHQ;B+ H /Bb+Qm`b2b U2bT2+B HHv BM
K Mv Q7 Qm` BMi2`pB2r22b KQ`2 +Q;MBx Mi Q7 i?2 `BbFb #2?BM/ i2
zQ`/BM; i?2K KQ`2 # H M+2/ QmiHQQFX JQ`2Qp2`- BKKB;` iBQM b
;HQ# H KQp2K2MiiQ/ v Bb i` MbM iBQM H T`Q+2bb- r?2`2 i?Qb2 BM `2
+mHim` H iB2b rBi? i?Qb2 r?Q `2K BM BM b2M/BM; bQ+B2iB2b pB p `
K2/B TH i7Q`Kb U" H/ bb ` 2i HX- kyRec G2pBii M/ C rQ`bFv- kyydc G2
kyRRVX "2+ mb2 r2 7QmM/ bi`QM; 2pB/2M+2 Q7 bm+? T`Q+2bb2b BM Q
Qm` T `iB+BT Mib `2 BM mMB[m2 TQbBiBQM iQ bT2 F #Qi? +`BiB+ H
i?2 bQ+BQHQ;B+ H mM/2`TBMMBM;b Q7 i?2B` H M;m ;2bX

Pp2`@b KTHBM; i2+? rQ`F2`b K v BMi`Q/m+2 MQi?2` 7Q`K Q7 b2H2
+ HHv- `2b2 `+? bm;;2bib i? i i2+? rQ`F2`b- ;Bp2M i?2B` BMbB/2` bi
i2+?MQ@QTiBKBbK TQbBiBp2 QmiHQQF rBi? `2bT2+i iQ i2+?MQHQ;B
kyReVX q?BH2 bm+? M 2z2+i + MMQi #2 /QrMTH v2/- i2+? rQ`F2`bö
i2+?MQHQ;B+ H T` +iB+2b +QmH/ BM 7 +i BK#m2 BM i?2K +`BiB+ H `2
BMiQX b T`QD2+i2/- r?BH2 K Mv T `iB+BT Mib bT2+mH i2/ QM i?2 #2M
r2`2 2[m HHv F22M QM mM/2`b+Q`BM; i?2 TQi2MiB H `BbFb bm+? M BM

i?2B` p2`v QrM H M;m ;2 +QKKmMBiB2bX h?2b2 Mm M+2/ T2`bT2+iBp2
`2b2 `+? T`Q+2bb2b M/ T`Q+2/m`2bX

kXRXR q?v b?QmH/ r2 T`BQ`BiBx2 HQr@`2bQm`+2 H M;m ;2b\

G M;m ;2 Bb MQi QMHv r v 7Q` T2QTH2 iQ +QKKmMB+ i2 rBi? QM2 MQ
+mHim`2- ?BbiQ`v- M/ b2H7@B/2MiBiv U.2KB+?2HBb M/ q2B#mHH- k
;2Mi- H M;m ;2 7Qbi2`b +QKKmMBiv #v 2ti2M/BM; i?2 i` /BiBQM M/
T2QTH2X 1p2M i?Qm;? K Mv Q7 Qm` HQr@`2bQm`+2 H M;m ;2 BMi2`pB
bT2 F2`b- HKQbi HH Q7 i?2K K BMi BM i? ii?2B` M iBp2 iQM;m2 `2K
Q7 i?2B` B/2MiBivX .` rBM; T ` HH2Hb #2ir22M i?2Kb2Hp2b M/ i?2B` M
i? M ? H7 Q7 i?2 T `iB+BT Mib Q7 Qm` bim/v H K2Mi i? irBi?QmibmbiB
i?2 mb ;2 M/ TTHB+ iBQM Q7 i?2B` M iBp2 H M;m ;2b- K Mv Q7 i?2K rQ
BM iBK2 iQ +QK2X

.2+HBM2 Q7 L iBp2 G M;m ;2 Mh?2mH2m`i2Xi i?2 HQr@`2bQm`+2 H M;r
KB;?i #2 mM/2`;QBM; bi i2 Q7 /2+HBM2 `2p2`#2` i2/ i?`Qm;?Qmi i?2
iBQMbivTB+ HHv ii`B#mi2/ i?2 /2+HBM2 iQ irQ + mb2b, +mHim` H M/2
bm;;2bib i? i b KQ`2 M/ KQ`2 bT2+ib Q7 Qm` HBp2b #2+QK2 /B;Bi H
2tTQbm`2b iQ +QMi2Mi 7QmM/ QM i?2 r2# M/ bQ+B H K2/B TH i7Q`Kb U
H2 /b iQ i?2 T`BQ`BiBx iBQM Q7 ?B;?@`2bQm`+2 H M;m ;2bX "v 2t
bTQiHB;?ib q2bi2`M 2TBbi2KQHQ;v M/ B/2 b Qp2` Qi?2` r vb Q7 FMQrB
72r BMi2`pB2r22b TQBMi2/ Qmi- i?2 +mHim` H /QKBM M+2 Q7 i?2 q2b
QMiQ KQ`2 HQ+ HBx2/ K2/B T`Q/m+iBQMbX b HQr@`2bQm`+2 H M;m ;
#QQFb- KQpB2b- M/ bQ+B H K2/B +QMi2Mi i BHQ`2/ iQ ?B;?@`2bQm`
2bi BM +QMi2Mi T`Q/m+2/ BM i?2B` M iBp2 iQM;m2 +QmH/ #2 +`Qr/2/
m/B2M+2b- +mHim` H T`Q/m+ib BM HQr@`2bQm`+2 H M;m ;2b `BbF /
MQi?2` + KT ii`B#mi2b HQr@`2bQm`+2 H M;m ;2bö /2+HBM2 iQ i?2
B+ H 2+QMQKvX 6Q` K Mv HQr@`2bQm`+2 H M;m ;2 bT2 F2`b r?Q +QK
?B;?@`2bQm`+2 H M;m ;2 HBF2 1M;HBb? Bb b22M b #Qi? p2?B+H
M/ bQ+B H KQ#BHBivXHSBMQm`BiQ7MM2 ;i?2Q# H 2+QMQKv K2 Mb /B`2
KQ`2 `2bQm`+2b i 1M;HBb? 2/m+ iBQM M/ i2i?2`BM; HQ+ H +QKKm
FMQrH2/;2 2+QMQKv Km+? Q7 Bi /`Bp2M #v i?2 /2K M/b Q7 i?2 q2biX
x2`Q@bmK H2Mb- K Mv BMi2`pB2r22b #2HB2p2 i? i i?2 T`QKQiBQM
BM+`2 bBM; T2`BT?2` HBx iBQM Q7 M iBp2 H M;m ;2b BM Tm#HB+ H
bi imb Q7 K Mv HQr@`2bQm`+2 H M;m ;2b `BbFb +QMiBMm2/ `2H2; iB
LQiBM; i?2b2 i`2M/b- K Mv HQr@`2bQm`+2 H M;m ;2 bT2 F2`b `2
i` MbH iBQM +QmH/ #2 +`BiB+ H iQQH BM T`QKQiBM; H M;m ;2 M/
M A;#Q bT2 F2` m`;2/- BKT`QpBM; K +?BM2 i` MbH iBQM 7Q` ?Bb M
KQ`2 T2QTH2 iQ T`Q/m+2 +mHim` H FMQrH2/;2 BM i? i H M;m ;2- H i2`
qBFBT2/B +QmH/ #2 pBi H TH i7Q`K i? i 2M #H2 Qi?2`b iQ H2 `M #
M/ T` +iB+2bX 1+?QBM; bm+? b2MiBK2Mib- MQi?2` BMi2`pB2r22 T
bm+? #B@/B`2+iBQM H H2 `MBM;- MQiBM; i? i ? pBM; i?2 #BHBiv iQ i`
MQi bT2 F i?2B` H M;m ;2 +QmH/ `2 / M/ mM/2`bi M/ qBFBT2/B `iB+H
r?B+? 7m`i?2` KQiBp i2b Qi?2` r`Bi2`b iQ r`Bi2 KQ`2X h?mb- #B@/B`2
BHHmKBM i2b i?2 BMi`B+ i2 `2H iBQMb?BT #2ir22M K +?BM2 i` MbH i

Bi T`QpB/2b M QTTQ`imMBiv iQ /Bb`mTi i?2 2Mi`2M+?BM; M im`2 Q7 c
/Bbb2KBM iBQMX h?2 +2Mi` HBiv Q7 qBFBT2/B BM i?2b2 biQ`B2b i2
i?2 rQ`H/öb KQbi 7`2[m2Mi2/ FMQrH2/;2@b? `BM; TQ`i H +QmH/ /22T
Qm`2zQ`iX

*Qp2` ;2 M/ Zm HBiv Q7 1tBbiBM; miQK iBq+?2`MMbF2i/B@kQXi
i` MbH iBQM +Qp2` ;2- KQbi HQr@`2bQm`+2 H M;m ;2 bT2 F2`b 2tT`2
i?2B``2bT2+iBp2 H M;m ;2b `2 bmTTQ`i2/ #v 2tBbiBM; bvbi2KbX 72
#2BM; BM+Hm/2/ #v +QKK2`+B HHv p BH #H2 b2`pB+2b K F2b i?2K
pBbB#BHBiv Q7 i?2B` H M;m ;2bX >Qr2p2`- bm+? b2MiBK2Mib `2 M
b2H2+i ;`QmT Q7 HQr@`2bQm`+2 H M;m ;2 bT2 F2`b- r?Qb2 H M;m ;
M/ p `B Mib- 7mHH +Qp2` ;2 `2K BMb H +FBM;X

6Q` BMbi M+2- JQ`Q++ M ` #B+ bT2 F2` b B/ i? i 7mHHv bmTTQ`iB
`2[mB`2b mb iQ i F2 i?2 p `BQmb 2t BMiQ #B+QHm i;mbQ B7bi r2 /Q MQi 2M
mT 7 pQ`BM; QM2 7Q`K Qp2` MQi?2`X h?Bb +QM+2`M bBKBH `Hv TTH
KmHiBTH2 b+`BTib UBX2X- " MD `- E Mm`B 2i+XVX "v 2t+Hm/BM; +2
T`QTTBM; KQ`2 r2HH@`2bQm`+2/ p `B Mib b i?2 ó/27 mHiô QTiBQM U
r2 MQi QMHv D2QT `/Bx2 ++m` i2 +mHim` H `2T`2b2Mi iBQM- #mi H
}2H/ i? i H`2 /v TH ;m2b H M;m ;2 /Bbi`B#miBQM M/ mb ;2 +`Qbb /Bz2
PM i?2 Qi?2`? M/- [m HBiv +QM+2`Mb `2bQM i2/ +`Qbb i?2 #Q `/ r
_2~2+iBM; QM bBx #H2 [m HBiv ; T TBi+?BM; ?B;?@`2bQm`+2 H M;m
`2bQm`+2 H M;m ;2 i` MbH iBQM UCQb?B 2i HX- kyRNV- K Mv BMi2`pB
`2bmHib b i?2 F2v `2 bQM 7Q` B``2;mH ` Q` /Bb+QMiBMm2/ mb2X 6Q`
b vb i? i i` MbH iBM; b2Mi2M+2 BM i?2B` H M;m ;2 mbBM; +QKK2`+B
i?2M 2/BiBM; Bi i F2b KQ`2 iBK2 i? M /QBM; bQ K Mm HHvX MQi?2` B
Bi Bb MQi T2`72+iBQM i? i b?2 r Mib- #mi ` i?2` i2+?MQHQ;v i? i BI
i` MbH iBQM iQ M/ 7`QK i?2B` H M;m ;2X 72r BMi2`pB2r22b 2p2M K2
+ `2 ;Bp2M iQ i?2B` H M;m ;2b BM bQK2 i` MbH iBQM TH i7Q`Kb ? p2
+`m/2 i` MbH iBQMb- 7m`i?2` 2`Q/BM; i?2B` +QM}/2M+2 BM i?2b2 bv
`2KBM/ mb i? i 2p2M i?Qm;? H M;m ;2 BM+HmbBQM Bb M BKTQ`i Mi}
M/ ?B;? [m HBiv i` MbH iBQM Bb biBHH r? i K ii2`b KQbi i i?2 2M/ Q7

q?Q bi M/b iQ ; BMb+mbbBQMb `QmM/ i?2 p Hm2 Q7 K +?BM2 i` MbH
`2bQm`+2 H M;m ;2 bT2 F2`b 2pBM+2 i?2 /22T bQ+BQ2+QMQKB+ ; Tb
7`QK MQi?2`- BKT +iBM; i?2 T2`+2Bp2/ miBHBiv Q7 i?2 i2+?MQHQ;vX
T`BK `BHv ?2HTb i?Qb2 7`QK KQ`2 /p Mi ;2/ # +F;`QmM/b H2 `M M2
KQ`2 2z2+iBp2Hv- Bib T`2b2M+2 BM }M M+B HHv BKTQp2`Bb?2/ +QKKr
7Q` bQ+B H KQ#BHBiv Q` 2p2M 2+QMQKB+ bm`pBp HX 6Q` BMbi M+2
BM 1i?BQTB - r?2`2 H2bb i? M ky T2`+2Mi Q7 i?2 +QmMi`v ? b BMi2`
iQ r? i i?2 r2# Qz2`b Bb 2p2M KQ`2 `2bi`B+i2/ /m2 iQ i?2 H +F Q7 [m H
H i2` bi`2bb2/ i? i H M;m ;2 + M #2 M 2Mi`2M+?2/ # ``B2` iQ 2/m+ i
J Mv HQr@`2bQm`+2 H M;m ;2 bT2 F2`b 7`QK 7`B+ 2+?Q i?2b2 b2M
i?2 +QMb2[m2M+2b Q7 +?`QMB+ K `;BM HBx iBQM M/ Bib BKT +i QM
M/ i?2 rB/2 bT2+i`mK Q7 ; BMb K +?BM2 i` MbH iBQM +QmH/ /2HBp2`

kX 6Q` /Bb+mbbBQM Q7 i?2 MQiBQM Q7 H M;mQB/- b22 :QQ/ M/ >2M/`vt@S `F2` U

wQQKBM; BMiQ BM/BpB/m H +QKKmMBiB2b i?2Kb2Hp2b-r2 b22 b
BMbi M+2-KQbi BMi2`pB2r22b ;`22 i? i i?Qb2 rBi? i2+?MQHQ;B+ H
KQ`2 7`QK K +?BM2 i` MbH iBQM i? M i?Qb2 r?Q /Q MQiX PM2 BMi2`
BM/BpB/m Hb BM i?2B` +QKKmMBiB2b `2 KQ`2 r2HH@bmBi2/ iQ 2t
i` MbH iBQM i? M i?2B` QH/2` +QmMi2`T `ibX *BiBM; i?2 `2+2Mi *P
2t KTH2-b?2 MQi2/ i? i BM TH +2b r?2`2 b+B2M+2@# +F2/ BM7Q`K i
H +F Q7 i`mbi@rQ`i?v 7Q`K H BMbiBimiBQMb-b2MBQ`b Q7 i?2b2 +Q
i?2B` KQ`2 i2+?@b ppv M2irQ`F M/ 7 KBHv K2K#2`b iQ +[mB`2 iBK
BM7Q`K iBQM /2`Bp2/ 7`QK BMi2`M iBQM H Q`; MBx iBQMbX AM i?2
H2p2Hb Q7 i2+?MQHQ;v FMQr@?Qr rQmH/ HbQ #2 #2ii2` #H2 iQ `2
M2rb-Q` QMHBM2 b+ Kb i? i+QmH/ `Bb2 7`QK i?2 2tT MbBQM Q7 i`
HQr@`2bQm`+2 H M;m ;2bX

h F2M +QHH2+iBp2Hv-Bi Bb BKTQ`i Mi iQ MQi2 i? i HQr@`2bQm`+2
MQi KQMQHBi?B+ ;`QmTc i?2v 2 +? M pB; i2 mMB[m2 bQ+BQTQHBiB
bT2 FBM; iQ i?2B` +QMbiBim2Mib-r2 H2 `M i? i`2 HBxBM; [m HBiv i` M
b2p2` H `2 bQMb-`2K BMb QM2 bQHmiBQM iQ K bbBp2 TmxxH2 i? i Bb
M/ 2[mBi #H2 FMQrH2/;2 ++2bbX h? i b B/-#v Qz2`BM; mT QM2 bQHm
Qi?2` +iQ`b BMiQ +iBQMX b QM2 HQr@`2bQm`+2 H M;m ;2 bT2 F2`
HQr@`2bQm`+2 H M;m ;2b BM K +?BM2 i` MbH iBQM ?2HTb /2@T`BQ
pBbB#BHBiv QM ;HQ# H b+ H2-r?B+? +QmH/ +QKT2H HQ+ H BMbiB
KQ`2 b2`BQmbHv M/ BMp2bi KQ`2 `2bQm`+2b BMiQ T`2b2`pBM; Q` i2 +
mM/2`b+Q`2b #Qi? i?2 bvK#QHB+ M/ K i2`B H #2M2}ib K +?BM2 i` Mb
TQbBiBp2 2M+Qm` ;2K2Mib 7`QK HQr@`2bQm`+2 H M;m ;2 bT2 F2`b i
bim/v `2KBM/ mb i? i#v i FBM; ?mK M@+2Mi`B+ TT`Q +? M/ 7Q+mb
? p2 ?BbiQ`B+ HHv #22M H27i#2?BM/-r2 + M ?2HT +QKKmMBiB2b K I
M iBp2 H M;m ;2b [mBMi2bb2MiB H T `i Q7 K Mv T2QTH2öb +mHim`2

kXk LQ G M;m ;2 G27i "2?BM/, :mB/BM; S`BM+BTH2b

*QK#BMBM; BMbB;?ib /` rM 7`QK BMi2`pB2rb rBi? HQr@`2bQm`+2 H
T` +iB+2b /BbiBHH2/ 7`QK HBi2` im`2 QM `2bTQMbB#H2 A U ``B2i 2i
"HQ/;2ii 2i HX-kykkc S mHH / 2i HX-kykRc a K# bBp M M/ >QH#`Q
7Qm` F2v ;mB/BM; T`BM+BTH2b mM/2`HvBM; Qm` `2b2 `+?,

RX S`BQ`BiBx2 i?2 M22/b Q7 mM/2`b2`p2/b+QQK2MiB2bM2/-r2 Tmi
i?2 M22/b Q7 HQr@`2bQm`+2 H M;m ;2 +QKKmMBiB2b ii?2 7`QMi
_2+Q;MBxBM; i? iK +?BM2 i` MbH iBQM Bb p Hm2@H /2M i2+?M
?BbiQ`B+ HHv /2@T`BQ`BiBx2/ +2`i BM TQTmH iBQMb-r2 mb2 i?B
M/ `2bQm`+2b iQ mM/2`b2`p2/ +QKKmMBiB2bX "v 2H2p iBM; i?2
H M;m ;2 +QKKmMBiB2b-r2 ?QT2 i? iQm` +QMi`B#miBQM Bb T `i Q
T`QT2Hb /B;Bi H `2T`2b2Mi iBQM BMiQ KQ`2 2[mBi #H2 2` X

kXa? `BM; i?`Qm;? QT2MX@bQm@+2MQm`+2 H M;m ;2 bT2 F2`b +`Qbb
`2KBM/mb i? ii` MbT `2M+v Qm;?i iQ #2 F2v 2KT? bBb r?2M /2p2H
i?2 /m H BMi2Mi iQ 7Qbi2` i` MbT `2M+v M/ pQB/ /mTHB+ iBQM Q
QM i? ir2 r2`2 ;QBM; iQ QT2M bQm`+2 LGG"X h?Bb r v-i?2 `2b2 `+?

+QmH/ /B`2+iHv #2M2}i 7`QK Qm` +QMi`B#miBQMX *`2 iBM; LGG
KBM/ HbQ KQiBp i2b mb iQ #2 BMi2MiBQM H M/ /2HB#2` iBp2 BM
i?2 /2p2HQTK2Mi H T`Q+2bbX q2 ?QT2 i? i i?2 BKT +i Q7 Qm` rQ`
b Qi?2` b+B2MiBbib M/ T` +iBiBQM2`b #mBH/ QM i?Bb 2zQ`i iQ /p
i` MbH iBQM b r?QH2X

jX"2BM; BMi2`/Bb+BTHBM `v BMX Qm+QT2MQHv?Tmi #v HQr@`2bQm
H M;m ;2 bT2 F2`-K +?BM2 i` MbH iBQM Bb MQi Dmbi +Q/BM; T`Q
BiBb ?mK MK ii2`X hQ pQB/i?2 õ HB;MK2Mi T`Q#H2Kö U*?`BbiB
bvbi2K iQ T2`7Q`K BM r v i? iBb #Qi? p Hm2@b2MbBiBp2 M/ bQ
`2b2 `+? 2zQ`iBb i F2M QM #v MBMi2`/Bb+BTHBM `v i2 KrBi? b+?
Q7 ?mK MBiB2b UBX2X-S?BHQbQT?v-1i?B+bV-bQ+B H b+B2MiB
M/ i2+?MB+ H UBX2X-*QKTmi2` a+B2M+2-ai iBbiB+bV # +F;`QmM
Q7 Qm` i2 K MQi QMHv 2tT M/b Qm` K2i?Q/QHQ;B+ H M/ M HviB+
mb +? M+2 iQ H2p2` ;2 /Bz2`2MibFBHHb iQ i +FH2 /BbT ` i2 bT2

9X"2BM; `2~2tBp2 BM QXm62xQHHv-`2~2tBpBiv KQiBp i2b mb iQ +`BiB
Qm` QrM Dm/;K2Mib-T` +iB+2b- M/ #2HB27 bvbi2Kb i?`Qm;?Qmi L
iQ 2Mbm`2 i? ir2 KBiB; i2 #B b2b +QKKQMHv 7QmM/ BM i?2 /2p2
BMi2HHB;2M+2 bvbi2KbX *QM+`2i2Hv-r2 Qz2` mT/2i BH2//Q+mK2
ip `BQmb/2+BbBQMb #2HQriQ HHQr/Bz2`2Mibi F2?QH/2`biQ +Q
M/ KQiBp iBQMbX q2 +FMQrH2/;2 i? irBi? K#BiBQmb 2zQ`ib H
? p2 iQ #2 K /2 M/ T2`72+iBQM `2K BMb 2HmbBp2X b bm+?-Bi
+m``2Mi 2zQ`irQmH/ BMpBi2 +`BiB+ H 2t KBM iBQMb Q7 2tBbiB
i?2M HHQr mb iQ K F2 KQ`2 BM7Q`K2//2+BbBQMb BM 7mim`2 Bi2

LQr i? ir2 ? p2/2b+`B#2/Qm` KQiBp iBQM M/ p Hm2b-r2 KQp2 QM iQ
biQ`v Qp2`+QKBM;i?2 i2+?MB+ H +? HH2M;2b BMpQHp2/BM `2 HBxBM
H M;m ;2b-7`QK H M;m ;2 B/2MiB}+ iBQM iQ i` BMBM;/i-KQ/2Hb-
+ b2rBi? Mv+miiBM;2/;2 BMi2`p2MiBQMb-#B;T`Q#H2Kb `2[mB`2 M
/2b+`B#2 i?2 DQm`M2v r2 iQQF iQ K i2`B HBx2 i?2 i2+?MB+ H /BK2M
2i?B+ H M/ bQ+B H +QMbB/2` iBQMb HQM;i?2 r vX 6B`bi-H2iöb K22

jX G M;m ;2b

"`Q /Hv ++2bbB#H2 K +?BM2 i` MbH iBQM bvbi2Kb bmTTQ`i `QmM
Bb iQ #`BM;i?Bb MmK#2` mT iQ kyyX AM /2+B/BM;r? i H M;m ;2b iQ
i?`Qm;? i?2 RyR H M;m ;2bQ+Qb2@2RyRkb2i7Q` i` MbH iBQM 2p Hm i
+Qp2`BM;T`2/QKBM MiHv HQr@`2bQm`+2 H M;m ;2bX 6`QK i?2`2-r2 ;
Q7 Qp2` k8y TQbbB#H2 H M;m ;2 + M/B/ i2b-2p2Mim HHv i`BKKBM;
}M H 2tT MbBQM 7`QK RyR iQ kyyY H M;m ;2bX

h?2 +`2 iBQM T`Q+2bb Q7 i?2 T`2HBKBM `v HBbi Bb b 7QHHQrb
H M;m ;2b rBi? qBFBT2/B T`2b2M+2X b MQi2/BM i?2 b2+iBQM #Qp2
Q7 FMQrH2/;2 /Bbb2KBM iBQM 7Q` K Mv bT2 FBM;HQr@`2bQm`+2 H M
TH +2 iQ bi `iX *m``2MiHv-qBFBT2/B bmTTQ`ib Qp2` jyy H M;m ;2
Bib +QMi2Mi #2vQM/1M;HBb? UCQ?MbQM M/G2b+F-kykkV-M/M2

*Q/2	G M;m ;2	a+`BTi	6 KBHv	am#;`QmTBM;V _2bX aT2+B}+ iBQM
+2n ` M2r	+2?M2b2	` #B+	mbi`QM2bB M	J H vQ@SQHvM2bB M LQ`i? +2?M2b2
+2nG iM	+2?M2b2	G iBM	mbi`QM2bB M	J H vQ@SQHvM2bB M LQ`i? +2?M2b2
+Kn ` #M2r	J2bQTQi KB M ` #B+	` #B+	7`Q@ bB iB+	a2KBiB+ 7 GQr " ;?/ /B
+[n ` #M2r	h BxxB@ /2MB ` #B+	` #B+	7`Q@ bB iB+	a2KBiB+ 7 GQr
2#n ` #M2r	hmMBbB M ` #B+	` #B+	7`Q@ bB iB+	a2KBiB+ 7 GQr .2`D
7`nG iM	7`BF Mb	G iBM	AM/Q@1m`QT2 M	:2`K MB+ V >B;?
DTn `#M2r	aQmi? G2p MiBM2 ` #B+	` #B+	7`Q@ bB iB+	a2KBiB+ 7 GQr KK MB
F nG iM2r	F M	G iBM	iH MiB+@*QM;Q Er oQHi @*QM;Q	7 GQr b Mi2
K?n1i?B	K? `B+	:2 2x	7`Q@ bB iB+	a2KBiB+ V GQr //Bb # #
T+n ` #M2r	LQ`i? G2p MiBM2 ` #B+	` #B+	7`Q@ bB iB+	a2KBiB+ 7 GQr
`#n ` #	JQ/2`M ai M/ `/ ` #B+	` #B+	7`Q@ bB iB+	a2KBiB+ V >B;?
`#nG iM2r	JQ/2`M ai M/ `/ ` #B+	G iBM	7`Q@ bB iB+	a2KBiB+ 7 GQr
`bn ` #M2r	L D/B ` #B+	` #B+	7`Q@ bB iB+	a2KBiB+ 7 GQr
`vn ` #M2r	JQ`Q++ M ` #B+	` #B+	7`Q@ bB iB+	a2KBiB+ 7 GQr
`xn ` #M2r	1;vTiB M ` #B+	` #B+	7`Q@ bB iB+	a2KBiB+ 7 GQr
bKn"2M;	bb K2b2	"2M; HB	AM/Q@1m`QT2 M	AM/Q@ `v M GQr 1 bi2`M
binG iM	bim`B M	G iBM	AM/Q@1m`QT2 M	Ai HB+ 7 GQr *2Mi` H
r n.2pM2r	r /?B	.2p M ; `B	AM/Q@1m`QT2 M	AM/Q@ `v 7 M GQr vQ/?v
v`nG iM2r	*2Mi` H vK `	G iBM	vK ` M	*2Mi` H aQmi?2`M vK`r GQr vK ` G S x DBH i
x#n ` #M2r	aQmi? x2`# BD MB	` #B+	hm`FB+	*QKKQM hm`FB+ 7 GQr h #`BxB
xDnG iM	LQ`i? x2`# BD MB	G iBM	hm`FB+	*QKKQM hm`FB+ V GQr a?B`p M
# Fn*v iM2r	" b?FB`	*v`BHHB+	hm`FB+	*QKKQM hm`FB+ GQr GBi2` `v
# KnG iM2r	" K# `	G iBM	J M/2	q2bi2`M J M/2 V GQr
# MnG iM2r	" HBM2b2	G iBM	mbi`QM2bB M	J H vQ@SQHvM2bB M 7 GQr
#2Hn*v`H	"2H `mbB M	*v`BHHB+	AM/Q@1m`QT2 M	" HiQ@aH pB+ GQr *2Mi` H
#2KnG iM2r	"2K#	G iBM	iH MiB+@*QM;Q	"2Mm2@*QM;Q 7 GQr *2Mi` H
#2Mn"2M;	"2M; HB	"2M; HB	AM/Q@1m`QT2 M	AM/Q@ `v M >B;? _ `?B
#?Qn.2p M2r	"?QDTm`B	.2p M ; `B	AM/Q@1m`QT2 M	AM/Q@ `v M GQr
#DMn `M2r#	" MD `	` #B+	mbi`QM2bB M	J H vQ@SQHvM2bB M 7 GQr MD `Em H
#DMnG iM2r	" MD `	G iBM	mbi`QM2bB M	J H vQ@SQHvM2bB M 7 GQr MD `Em H
#Q/nhB#M2r	ai M/ `/ hB#2i M	hB#2i M	aBMQ@hB#2i M	"Q/B+ V GQr G? b
#QbnG iM	"QbMB M	G iBM	AM/Q@1m`QT2 M	" HiQ@aHVpB+ >B;?
#m;nG iM2r	"m;BM2b2	G iBM	mbi`QM2bB M	J H vQ@SQHvM2bB M 7 GQr M"QM2
#mHn*v`H	"mH; `B M	*v`BHHB+	AM/Q@1m`QT2 M	" HiQ@aH pB+ >B;?
+ inG iM	* i H M	G iBM	AM/Q@1m`QT2 M	Ai HB+ V >B;?
+2#nG iM	*2#m MQ	G iBM	mbi`QM2bB M	J H vQ@SQHvM2bB M 7 GQr
+2bnG iM	*x2+?	G iBM	AM/Q@1m`QT2 M	" HiQ@aHVpB+ >B;?
+DFnG iM2r	*?QFr2	G iBM	iH MiB+@*QM;Q	"2Mm2@*QM;Q 7 GQr
+F#n ` #	*2Mi` H Em`/Bb?	` #B+	AM/Q@1m`QT2 M	A` MB M V GQr
+`?nG iM2r	*`BK2 M h i `	G iBM	hm`FB+	*QKKQM hm`FB+ 7 GQr
+vKnG iM	q2Hb?	G iBM	AM/Q@1m`QT2 M	*2HiB+ V GQr u qvM/Q/2;
/ MnG iM	. MBb?	G iBM	AM/Q@1m`QT2 M	:2`K MB+ V >B;?
/2mnG iM	:2`K M	G iBM	AM/Q@1m`QT2 M	:2`K MB+ V >B;?
/BFnG iM2r	aQmi?r2bi2`M .BMF	G iBM	LBHQiB+	q2bi2`M LBHQiB+ 7 GQr _2F
/vmnG iM2r	.vmH	G iBM	J M/2	q2bi2`M J M/2 7 GQr
/xQnhB#M2r	.xQM;F?	hB#2i M	aBMQ@hB#2i M	"Q/B+ 7 GQr
2HHn:`2F	:`22F	:`22F	AM/Q@1m`QT2 M	:` 2+Q@S?Vv;B+ >B;?
2M;nG iM	1M;HBb?	G iBM	AM/Q@1m`QT2 M	:2`K MB+ V >B;?
2TQnG iM2r	1bT2` MiQ	G iBM	*QMbi`m+i2/	1bT2` MiB+ V GQr
2binG iM	1biQMB M	G iBM	l` HB+	6BMMB+ V >B;?
2mbnG iM2r	" b[m2	G iBM	" b[m2	V >B;?
2r2nG iM2r	1r2	G iBM	iH MiB+@*QM;Q	Er oQHi @*QM;Q 7 GQr HQ
7 QnG iM2r	6 `Q2b2	G iBM	AM/Q@1m`QT2 M	:2`K MB+ V GQr
}DnG iM2r	6BDB M	G iBM	mbi`QM2bB M	J H vQ@SQHvM2bB M 7 GQr M" m
}MnG iM	6BMMBb?	G iBM	l` HB+	6BMMB+ V >B;?
7QMnG iM2r	6QM	G iBM	iH MiB+@*QM;Q	Er oQHi @*QM;Q 7 GQr
7` nG iM	6`2M+?	G iBM	AM/Q@1m`QT2 M	Ai HB+ V >B;?
7m`nG iM2r	6`BmHB M	G iBM	AM/Q@1m`QT2 M	Ai HB+ 7 GQr *2Mi` H
7mpnG iM	LB;2`B M 6mH7mH/2	G iBM	iH MiB+@*QM;Q	LQ`i?@*2Mi` H GQH MaBQ+FQiQ
;H nG iM2r	a+QiiBb? : 2HB+	G iBM	AM/Q@1m`QT2 M	*2HiB+ 7 GQr LQ`i?2`M >2#`B/2b

*Q/2	G M;m ;2	a+`BTi	6 KBHv	am#;`QmTBM_;2bX	aT2+B}+ iBQM
;H2nG iM	A`Bb?	G iBM	AM/Q@1m`QT2 M	*2HiBV+ GQr	
;H;nG iM	: HB+B M	G iBM	AM/Q@1m`QT2 M	Ai HBV+ GQr	
;`MnG iM	:m ` MB	G iBM	hmTB M	J r2iB@:mV` MBQr	
;mDn:mD`	:mD ` iB	:mD ` iB	AM/Q@1m`QT2 M	AM/QV@ `vGMr	K/ p /Bfam`iB
? inG iM	> BiB M *`2QH2	G iBM	AM/Q@1m`QT2 M	Ai HBV+ GQr	
? mnG iM	> mb	G iBM	7`Q@ bB iB+	*? /B+ V GQr	
?2#n>2#`	>2#`2r	>2#`2r	7`Q@ bB iB+	a2KBiB+V >B;?	
?BMn.2p	>BM/B	.2p M ; `B	AM/Q@1m`QT2 M	AM/QV@ `vGMr	
?M2n.2p	*?? iiBb; `?B	.2p M ; `B	AM/Q@1m`QT2 M	AM/Q7@ `vGMr	
?`pnG iM	*`Q iB M	G iBM	AM/Q@1m`QT2 M	" HiQV@aH+Bb;B?+	
?mMnG iM	>mM; `B M	G iBM	l` HB+	V >B;?	
?v2n `KM	`K2MB M	`K2MB M	AM/Q@1m`QT2 M	`K2MB+GQr	u2`2p M
B#QnG iM	A;#Q	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ	*2Mi` H
BHQnG iM	AHQ+ MQ	G iBM	mbi`QM2bB M	J H vQV@SQGHQvM2bB M	
BM/nG iM	AM/QM2bB M	G iBM	mbi`QM2bB M	J H vQV@SQ>BvM2bB M	
BbHnG iM	A+2H M/B+	G iBM	AM/Q@1m`QT2 M	:2`K MB+>B;?	
Bi nG iM	Ai HB M	G iBM	AM/Q@1m`QT2 M	Ai HBV+ >B;?	
D pnG iM	C p M2b2	G iBM	mbi`QM2bB M	J H vQV@SQGHQvM2bB M	
DTMnCT M	C T M2b2	C T M2b2	C TQMB+	C T M2bB+>B;?	
F #nG iM	E #vH2	G iBM	7`Q@ bB iB+	"2`#2`7 GQr	LQ`i?1bi2`M
F +nG iM	CBM;T?Q	G iBM	aBMQ@hB#2i M	"` ?K7TmiGQr	
F KnG iM	E K#	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ	J +? FQb
F MnEM/	E MM /	E MM /	.` pB/B M aQmi?.` pB/BGMQr		*2Mi` H
F bn ` #nG	E b?KB`B	` #B+	AM/Q@1m`QT2 M	AM/Q@ `vGMr	EBb?ir `B
F bn.2p	E b?KB`B	.2p M ; `B	AM/Q@1m`QT2 M	AM/Q7@ `vGMr	EBb?ir `B
F in:2Q`	:2Q`;B M	:2Q`;B M	E `ip2HB M	:2Q`;B MV@wGMQr	E `iHB M
FM+n `#nG	*2Mi` H E Mm`B	` #B+	a ? ` M	q2bi2`M a ?7` MGQr	u2`r
FM+nGiM	*2Mi` H E Mm`B	G iBM	a ? ` M	q2bi2`M a ?7` MGQr	u2`r
F xn*v`H	E x F?	*v`BHHB+	hm`FB+	*QKKQM hm>BF;B?+	
F#TnGiM	E #Bv	G iBM	iH MiB+@*QM;Q	LQ`i?bQHGQr*QMQ;rQ	
F2 nG iM	E #mp2`/B Mm	G iBM	AM/Q@1m`QT2 M	Ai HB7+ GQr	aQi p2MiQ
F?KnE?K`	E?K2`	E?K2`	mbi`Q bB iB+	E?K2`B+V GQr	*2Mi` H
FBFnGiM	EBFmvm	G iBM	iH MiB+@*QM;Q	"2Mm27@*QGMQ;rQ	aQmi?2`M
FBMnGiM	EBMv `r M/	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ	
FB`n*v`H	Ev`;vx	*v`BHHB+	hm`FB+	*QKKQM hmGFB+	LQ`i?2`M
FK#nG iM	EBK#mM/m	G iBM	iH MiB+@*QM;Q	"2Mm27@*QGMQ;rQ	
FK`nG iM	LQ`i?2`M Em`/Bb?	G iBM	AM/Q@1m`QT2 M	A` MBVM GQr	
FQMnGiM	EBFQM;Q	G iBM	iH MiB+@*QM;Q	"2Mm27@*QGMQ;rQ	
FQ`n> M;	EQ`2 M	> M;mH	EQ`2 MB+	EQ`2 M V >B;?	
H QnG QQ	G Q	G Q	h B@E / B	E K@h B V GQr	oB2MiB M2
HBDnGiM	GB;m`B M	G iBM	AM/Q@1m`QT2 M	Ai HB7+ GQr	w2M2Bb2
HBKnGiM	GBK#m`;Bb?	G iBM	AM/Q@1m`QT2 M	:2`K MB+GQr	J bi`B+?iB M
HBMnG iM	GBM; H	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ	
HBinG iM	GBi?m MB M	G iBM	AM/Q@1m`QT2 M	" HiQV@aH+Bb;B?+	
HKQnGiM	GQK# `/	G iBM	AM/Q@1m`QT2 M	Ai HB7+ GQr	q2bi2`M
Hi;nG iM	G i; HB M	G iBM	AM/Q@1m`QT2 M	" HiQ7@aHGQB+	*2Mi` H
HixnG iM	Gmt2K#Qm`;Bb?	G iBM	AM/Q@1m`QT2 M	:2`K MB+GQr	
Hm nGiM	Gm# @E b B	G iBM	iH MiB+@*QM;Q	"2Mm27@*QGMQ;rQ	
Hm;nG iM	: M/	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ	
HmQnG iM	GmQ	G iBM	LBHQiB+	q2bi2`M LBHQQB+	
HmbnGiM	JBxQ	G iBM	aBMQ@hB#2i M	EmFBV@*?BMQr@L ;Bx rH	
HpbnG iM	ai M/ `/ G ipB MG	iBM	AM/Q@1m`QT2 M	" HiQV@aH+Bb;B?+	
K ;n.2pM	J ; ?B	.2p M ; `B	AM/Q@1m`QT2 M	AM/Q7@ `vGMr	: v
K Bn.2pM	J Bi?BHB	.2p M ; `B	AM/Q@1m`QT2 M	AM/QV@ `vGMr	
K HnJHvK	J H v H K	J H v H K	.` pB/B M aQmi?.` pB/BGMQr		
K `n.2p	J ` i?B	.2p M ; `B	AM/Q@1m`QT2 M	AM/QV@ `vGMr	o `? /B
KBMn `#nG	JBM M;F # m	` #B+	mbi`QM2bB M	J H vQ@SQGHQvM2bB M	
KBMnGiM	JBM M;F # m	G iBM	mbi`QM2bB M	J H vQ@SQGHQvM2bB M	
KF/n*v`H	J +2/QMB M	*v`BHHB+	AM/Q@1m`QT2 M	" HVQ@>aBH?pB+	
THinGiM	SH i2 m J H ; bv	G iBM	mbi`QM2bB M	J H vQV@SQGHQvM2bB M	
KHinG iM	J Hi2b2	G iBM	7`Q@ bB iB+	a2KBiBV+ >B;?	

*Q/2	G M;m ;2	a+`BTi	6 KBHv	am#;`QmTBM_;2bX aT2+B}+ iBQM
KMBn"2M;	J2Bi2B	"2M; HB	aBMQ@hB#2i M	EmFB7@*?GBQr@L ;
F?Fn*v`H	> H? JQM;QHB M	*v`BHHB+	JQM;QHB+@E?Bi M	JQM;QHB+
KQbnG iM	JQbbB	G iBM	iH MiB+@*QM;Q	LQ`i?7oQHG@r*QM;Q
K`BnG iM	J Q`B	G iBM	mbi`QM2bB M	J H vQ@SQGHQrM2bB M
Kv nJvK`	"m`K2b2	Jv MK `	aBMQ@hB#2"imMQ@ZB M;B+	V GQr J M/ H v@u M;QM
MH/nG iM	.mi+?	G iBM	AM/Q@1m`QT2 M	:2`K MB+>B;?
MMQnG iM	LQ`r2;B M LvMQ`bG	G iBM	AM/Q@1m`QT2 M	:2`K MB+GQr
MQ#nG iM	LQ`r2;B M "QFK´HG	G iBM	AM/Q@1m`QT2 M	:2`K MB+GQr
MTBn.2p	L2T HB	.2p M ; `B	AM/Q@1m`QT2 M	AM/QV@ `vGMQr 1 bi2`M
MbQnG iM	LQ`i?2`M aQi?Q	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ
MmbnG iM	Lm2`	G iBM	LBHQiB+	q2bi2`M LBHQQB+
Mv nG iM	Lv MD	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ
Q+BnG iM	P++Bi M	G iBM	AM/Q@1m`QT2 M	Ai HB7+ GQr
; xnG iM	q2bi*2Mi` H P`QKQ	G iBM	7`Q@ bB iB+	*mb?BiB+ GQr
Q`vnP`v	P/B	P`Bv	AM/Q@1m`QT2 M	AM/QV@ `vGMQr " H2br `B ULQ`i?2`MV
T ;nG iM	S M; bBM M	G iBM	mbi`QM2bB M	J H vQ@SQGHQrM2bB M
T Mn:m`m	1 bi2`M S MD #B	:m`KmF?B	AM/Q@1m`QT2 M	AM/QV@ `vGMQr J D?B
T TnG iM	S TB K2MiQ	G iBM	AM/Q@1m`QT2 M	Ai HB7+ GQr _ºK2`@J /m`Q@CQMBb
T2bn ` #	q2bi2`M S2`bB M	` #B+	AM/Q@1m`QT2 M	A` MBVM >B;?
TQHnG iM	SQHBb?	G iBM	AM/Q@1m`QT2 M	" HiQV@aH>BpB?+
TQ`nG iM	SQ`im;m2b2	G iBM	AM/Q@1m`QT2 M	Ai HBV+ >B;? "` xBH
T`bn ` #	. `B	` #B+	AM/Q@1m`QT2 M	A` MBVM GQr E #mHB
T#in ` #	aQmi?2`M S b?iQ	` #B+	AM/Q@1m`QT2 M	A` MBVM GQr GBi2` `v
[mvnG iM	v +m+?Q Zm2+?m	G iBM	Zm2+?m M	*?BM+? vV GQr aQmi?2`M Zm2+?m
`QMnG iM	_QK MB M	G iBM	AM/Q@1m`QT2 M	Ai HBV+ >B;?
`mMnG iM	_mM/B	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ
`mbn*v`H	_mbbB M	*v`BHHB+	AM/Q@1m`QT2 M	" HiQV@aH>BpB?+
b ;nG iM	a M;Q	G iBM	iH MiB+@*QM;Q	LQ`i?7oQHG@r*QM;Q
b Mn.2p	a MbF`Bi	.2p M ; `B	AM/Q@1m`QT2 M	AM/QV@ `vGMQr
b inPH+F	a Mi HB	PH*?BFB	mbi`Q bB iB+	JmM/ B+ GQr
b+MnG iM	aB+BHB M	G iBM	AM/Q@1m`QT2 M	Ai HB7+ GQr GBi2` `v aB+BHB M
b?MnJvK	a? M	Jv MK `	h B@E / B	E K@h B 7 GQr
bBMnaBM?	aBM? H	aBM? H	AM/Q@1m`QT2 M	AM/QV@ `vGMQr
bHFnG iM	aHQp F	G iBM	AM/Q@1m`QT2 M	" HiQV@aH>BpB?+
bHpnG iM	aHQp2MB M	G iBM	AM/Q@1m`QT2 M	" HiQV@aH>BpB?+
bKQnG iM	a KQ M	G iBM	mbi`QM2bB M	J H vQ@SQGHQrM2bB M
bM nG iM	a?QM	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ
bM/n ` #	aBM/?B	` #B+	AM/Q@1m`QT2 M	AM/QV@ `vGMQr oB+?QHB
bQKnG iM	aQK HB	G iBM	7`Q@ bB iB+	*mb?BiB+ GQr LbQK
bQinG iM	aQmi?2`M aQi?Q	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ
bT nG iM	aT MBb?	G iBM	AM/Q@1m`QT2 M	Ai HBV+ >B;? G iBM K2`B+ M
HbnG iM	hQbF H# MB M	G iBM	AM/Q@1m`QT2 M	H# MB M>B;?
b`/nG iM	a `/BMB M	G iBM	AM/Q@1m`QT2 M	Ai HB7+ GQr GQ;m/Q`2b2 M/ * KTB/ M2b2
b`Tn*v`H	a2`#B M	*v`BHHB+	AM/Q@1m`QT2 M	" HiQV@aH>BpB+
bbrnG iM	ar iB	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ
bmMnG iM	amM/ M2b2	G iBM	mbi`QM2bB M	J H vQ@SQGHQrM2bB M
br2nG iM	ar2/Bb?	G iBM	AM/Q@1m`QT2 M	:2`K MB+>B;?
br?nG iM	ar ?BHB	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ EBmM;mD
bxHnG iM	aBH2bB M	G iBM	AM/Q@1m`QT2 M	" HiQ7@aH>BpB+
i Knh KH	h KBH	h KBH	.` pB/B M	aQmi? .` pB/BGMQr *?2MM B
i in*v`H	h i `	*v`BHHB+	hm`FB+	*QKKQM hmGFB+*2Mi` H M/ JB//H2
i2Hnh2Hm	h2Hm;m	h2Hm;m	.` pB/B M	aQmi? .` pB/BGMQr *Q bi H
i;Fn*v`H	h DBF	*v`BHHB+	AM/Q@1m`QT2 M	A` MB GMQr
i;HnG iM	h ; HQ;	G iBM	mbi`QM2bB M	J H vQ@SQ>BvM2bB M
i? nh? B	h? B	h? B	h B@E / B	E K@h B V >B;?
iB`n1i?B	hB;`BMv	:2•2x	7`Q@ bB iB+	a2KBiB+V GQr
i [nG iM	h K b?2[	G iBM	7`Q@ bB iB+	"2`#2` 7 GQr E H Mb `
i [nh7M	h K b?2[	hB}M ;?	7`Q@ bB iB+	"2`#2` 7 GQr E H Mb `
iTBnG iM	hQF SBbBM	G iBM	AM/Q@1m`QT2 M	:2`K MB+GQr
ibMnG iM	hbr M	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ a2?m`mib?2
ibQnG iM	hbQM;	G iBM	iH MiB+@*QM;Q	"2Mm2V@*QGMQ;rQ

*Q/2	G M;m ;2	a+`BTi	6 KBHv	am#;`QmTBWM; _2bX	aT2+B}+ iBQM
imFnG iM	hm`FK2M	G iBM	hm`FB+	*QKKQM hm`FB+GQr	h2F2
imKnG iM	hmK#mF	G iBM	iH MiB+@*QM;Q	"2Mm2@*QM;GQr	_mKT?B
im`nG iM	hm`FBb?	G iBM	hm`FB+	*QKKQM hm`FB+>B;?	
irBnG iM	hrB	G iBM	iH MiB+@*QM;Q	Er oQHiV@*QM;rQ	Fm T2K
ixKnh7M;	*2Mi` H iH b h K xB;?iB}M ;?		7`Q@ bB iB+	"2`#2` 7 GQr	
mB;n `#	lv;?m`	` #B+	hm`FB+	*QKKQM hm`FB+GQr	
mF`n*v`H	lF` BMB M	*v`BHHB+	AM/Q@1m`QT2 M	" HiQV@aH+bB?+	
mK#nG iM	lK#mM/m	G iBM	iH MiB+@*QM;Q	"2Mm2@*QM;GQr	
m`/n ` #	l`/m	` #B+	AM/Q@1m`QT2 M	AM/Q@ `v MGQr	G b?F `B
mxMnG iM	LQ`i?2`M lx#2F	G iBM	hm`FB+	*QKKQM hm`FB+>B;?	
p2+nG iM	o2M2iB M	G iBM	AM/Q@1m`QT2 M	Ai HB+ 7 GQr	o2MB+2
pB2nG iM	oB2iM K2b2	G iBM	mbi`Q bB iB+	oB2iB+ V >B;?	
r `nG iM	q ` v	G iBM	mbi`QM2bB M	J H vQ@SQHvQrbBhM+HQ# M	
rQHnG iM	qQHQ7	G iBM	iH MiB+@*QM;Q	LQ`i?@*2Mi`GHQriH.MFB+`	
t?QnG iM	s?Qb	G iBM	iH MiB+@*QM;Q	"2Mm2@*QM>Q;?	L;[BF
v//n>2#	1 bi2`M uB//Bb?	>2#`2r	AM/Q@1m`QT2 M	:2`K MB+V GQr	> bB/B+
vQ`nG iM	uQ`m#	G iBM	iH MiB+@*QM;Q	"2Mm2@*QM;GQr	»v¼ M/A#/M
vm2n> Mf	um2*?BM2b2	> M Uh` /BiBQM HV	aBMQ@hB#2i M	aBMVBiB>B;?	
x?Qn> Mb	*?BM2b2	> M UaBKTHB}2/V	aBMQ@hB#2i M	aBMVBiB>B;?	
x?Qn> Mi	*?BM2b2	> M Uh` /BiBQM HV	aBMQ@hB#2i M	aBMVBiB>B;?	
xbKnG iM	ai M/ `/J H v	G iBM	mbi`QM2bB M	J H vQ@SQHv>M2bB M	
xmHnG iM	wmHm	G iBM	iH MiB+@*QM;Q	"2Mm2@*QM>Q;?	

h #H2 Ry9 G M;m ;2b Q7 LQ G M;m ;2 G2q2i/B2b?THMv,i?2 H*QM/;2mH;Bb2M;m ;2 M+KB2Ti
M/H M;m6 K2BXHvh?2 bvkV#BQM/B+ i2bK +?BM2i` MbH iBQM bmTTQ`i#v:QQ7;H2 M
BM/B+ i2b bmTTQ`i_#vbBM2/Bi?2i2Xb B7 r2 +H bbB7v i?2 H M;m ;2aTb2?B}+ QBQHMQr@
+QMi BMb-B7 p BH #H2- //BiBQM H BM7Q`K iBQM6QHMQ i2 2@kMyzmb;2Tp2M2B+MB+
BM/B+ i2b M2r H M;m 6;2HQ`2/b2@+QykT `2/6HQQ`2b@XRyR

b T `i Q7 +QKKmMBiv `2j[m2bi-T2Qb+Q2HBBX+Bi2/HBbib Q7 H M;m ;2b b
p `BQmb `2;BQMb #v M iBp2 bT2 F2`b-7Q+mbBM; T `iB+mH `Hv QM 7`
H M;m ;2b i? i? p2 ?BbiQ`B+ HHv #22M mM/2``2T`2b2Mi2/BM i` MbH
kykyVX q2 i?2M 2t KBM2/H M;m ;2 +Qp2` ;2 BM KmHiBTH2 2tBbiBM;
H M;m ;2 T`Q+2bbBM; +QKKmMBiv- T vBM; 7Q+mb2/ ii2MiBQM QM
++QKT MvBM; 2p Hm iBQM / i b2ibX 6BM HHv- r2 +QMbB/2`2/i?2 /
H M;m ;2 #v HQQFBM; ii?2 TT`QtBK i2 MmK#2` Q7 M iBp2 bT2 F2`b M
p `B #H2b `2H2p MiiQ Qm` rQ`FX

L2ti- 7Q` 2 +? Q7 i?2 H M;m ;2 + M/B/ i2b- r2 T `iM2`2/ rBi? HBM
bT2+B HBx2/H M;m ;2 b2`pB+2 T`QpB/2`b iQ mM/2`bi M/B7 2 +? Q7
/ `/Bx2/ r`Bii2M 7Q`KX q2 /B/ i?Bb #2+ mb2 ? pBM; `2HB #H2- ?B
/ i b2iBb +`BiB+ H iQ ++2H2` i2/2tT2`BK2Mi H T`Q;`2bbX >Qr2p2`- T
7 B`Hv bi M/ `/Bx2/ r`Bii2M 7Q`Kb ? b MQi #H2 /QrMbB/2b Ub22 TT2
H M;m ;2b ? p2 M im` H p `B iBQMb M/ `2 #2BM; r`Bii2M BM /Bz2`2M
/Bz2`2Mi `2;BQMbX 6Q` BMbi M+2- H M;m ;2b bm+? b 6mH ? BM+Hm/2
H M;m ;2b bm+? b E b?KB`B M/ *2Mi` H E Mm`B +QMi BM KmHiBTH
avbi2K iB+ HHv /Q+mK2MiBM; i?2b2 /BK2MbBQMb ?2HT2/mb bb2bb
KmHiBTH2 p `B Mib Q7 /Bz2`2Mi H M;m ;2b Ubm+? b H M;m ;2b rBi? K
M im` H p `B iBQMVX

jX?iiTb,ffK2i XrBFBK2/B XQ`;frBFBfG M;m ;2nT`QTQb HnTQHB+v

AM i M/2K rBi? i?2b2 +QMbB/2` iBQMb- /2+B/BM; r?B+? H M;m ;2b iQ
mHiBK i2Hv + K2 /QrM iQ bb2bbBM; i?2 TQi2MiB H BKT +i r2 KB;?i ?
HQr@`2bQm`+2 H M;m ;2 +QKKmMBiB2bX 6Q` BMbi M+2- r2 2t+Hm/
HQr MmK#2` Q7 M iBp2 bT2 F2`bX qBi?Qmi +QM+2`i2/ TH M iQ i?Q
M22/b Q7 i?2b2 +QKKmMBiB2b M/ TQi2MiB H `BbFb r2 +QmH/ + mb2-
BM+Hm/BM; i?2B` H M;m ;2b BM Qm` 2zQ`iX E22TBM; BM HBM2 rBi? Qr
i?2 H M;m ;2b i? iK /2 i?2 }M H +mi? p2 T`2b2M+2 QM qBFBT2/B M/
mM/2``2T`2b2Mi2/ `2;BQMbX G bi #mi MQi H2 bi- Bi Bb rQ`i? MQiBM; i?
K Mv H M;m ;2b i? i /Q MQi? p2 r`Bii2M bi M/ `/b Q` `2 T`2/QKBM M
?QT2 i? i 7mim`2 `2b2 `+? +QmH/ /B`2+i KQ`2 ii2MiBQM i H M;m ;2b r

G M;m ;2 AM7Q`K AiBMQM+XQ`/ M+2 rBi? i?2 O"2M/2`_mH2 U"2M/2`- kyR
K `Bx2 BM7Q`K iBQM #Qmi 2 +? Q7 Qm` ky9 bmTTQ`i2/ H M;m ;2b BM

*Q/2Xq2 `2T`2b2Mi2 +? H M;m ;2 rBi? "*S 9d i ; b2[m2M+2 mbBM; i
ejN@j +Q/2 b i?2 # b2 bm#i ;- r?B+? r2 +QKTH2K2MirBi? AaP R8Nk9 b
+QHH2+i2/ `2bQm`+2b 7Q` b2p2` H H M;m ;2b BM KQ`2 i? M QM2 b+`B

G M;m ;2hX? 2`2 K v #2 KmHiBTH2 r vb iQ `272`iQ i?2 b K2 H M;m ;2c /m
HBKBi iBQMb- QMHv QM2 Q7 i?2 p2`bBQMb Bb /BbTH v2/X h?2 H M;
`272`2M+2/ rBi? K DQ` HBM;mBbiB+ BM7Q`K iBQM TH i7Q`Kb bm+?
M/ :HQiiQHQ; U> KK `bi`ºK 2i HX- kykkVX

a+`BTihX? 2 1M;HBb? M K2 Q7 i?2 b+`BTi Bb T`QpB/2/X b bQK2 H M;m
KQ`2 i? M QM2 b+`BTi- r2 rQ`F iQr `/b bmTTQ`iBM; i?Bb M im` H p `B

6 KBHv M/ am#;`QmqT2BTM;QXpB/2 G M;m ;2 7 KBHv BM7Q`K iBQM 7Q` 2
b2/ QM i?2 :HQiiQHQ; / i # b2 U> KK `bi`ºK 2i HX- kykkVX

q2# amTTQ`iqX 2t KBM2 B7 2 +? H M;m ;2 Bb bmTTQ`i2M#vQ:`QQ;H2 h`
JB+`QbQ7i h`8 M?2Hbi2KXWQBHM/B+ i2b i? i 2Bi?2` Q` #Qi? TH i7Q`Kb bmT
H M;m ;2X h?2 7bBMK#BQ+Hi2b i? i M2Bi?2` TH i7Q`K b9mTTQ`ib i?2 H M;m

_2bQm`+2@G2p2Hq2U+_2i2VQX`Bx2 H M;Qmr@2`2bQmi?2 `2 72r2` i? M
RJ Tm#HB+Hv p BH #H2- /2@/mTHB+ i2/ #Bi2tib KTH2brBi? Mv Qi?2
Q7 kyy H M;m ;2bX LQi2 i?Bb ;Q2b #2vQM/ +QmMiBM; 1M;HBb?@+2M
H M;m ;2b K v ? p2 p BH #H2 / i b2ib BM H M;m ;2b bTQF2M KQ`2 T`QK
6Q` 2t KTH2- K Mv +QmMi`B2b BM 7`B+ `2 6` M+QT?QM2X h?Bb `
+H bbB}2/ b HQr@`2bQm`+2X

aT2+B}+ iBHQ?MBXb +QHmKM +QMi BMb- B7 p BH #H2- //BiBQM H BM7
bT2+B}+ H M;m ;2 p `B2iv Q` `2;BQM `2T`2b2Mi2/X

h?2 H M;m ;2 BM7Q`K iBQM T`QpB/2/ BM h #H2 R `2~2+ib i?2 `2bQm`
6HQ`2b@+QyH2+iBQM 2zQ`ib- r?B+? `2 /2b+`B#2/ BM i?2 M2tib2+iBQ

9X?iiTb,ffi` MbH i2X;QQ;H2X+QKf
8X?iiT,ffrrrX#BM;X+QKfi` MbH iQ`
eX AM7Q`K iBQM r b ++2bb2/ QM CmM2 R8- kykk

6B;m`2>jK M@h` MbH i2/ . i b2i *QMi`B#miBQMb Q7 LQ G,M;m ;2 G27
b ?B;?HB;?i2/- i?2b2 / i b2ib 2M #H2 KQ/2H i` BMBM; M/ 2p Hm iBQ

9X *`2 iBM; S`Q72bbBQM HHv h` MbH i2/ . i b2ib, 6GP_1a@kyy
LGG"@a22/

GQr@`2bQm`+2 i` MbH iBQM 7 +2b b2p2` H +? HH2M;2b- }`bi M/ 7Q`2
AM i?Bb b2+iBQM- r2 /2b+`B#2 i?`22 +QKTQM2Mib /2pBb2/ iQ Qp2`+Q
6B;m`2 jX 6B`bi- r2 /2b+`B#2 i?`22 +QKTQM2Mib /2pBb2/ iQ Qp2`+Q
#2M+?K `F / i b2ii? i/Qm#H2bi?2 H M;m ;2 +Qp2` ;2 Q7 Q`2 2pBQmb 2z
RyXR h?2M- r2 i` +2 i?2 /2p2HQTK2Mi T`Q+2bb Q7 Q`2/Q#B2 2biBQM HHv@i`
jN HQr@`2bQm`+2 H M;m ;2b- ;BpBM; mb i?2 #BHBiv iQ i` BM Mv K
/ i X 6BM HHv- r2 /2b+`B#2/ i b2i BM KmHiBTH2 /Bz2`2Mi /QK BMb iQ
;2M2` HBx #H2 i` MbH iBQM + T #BHBivX h?2b2 `2bQm`+2b 2M #H2 i
KQ/2Hb 7Q` H M;m ;2b i? i T`2pBQmbHv ? / K `;BM H bmTTQ`iX

9XR 6GP_1a@kyy

K DQ` `2 Q7 7Q+mb BM K +?BM2i` MbH iBQM `2b2 `+? ? b #22M QM
[m HBiv 2p Hm iBQM / i b2ib- Q` #2M+?K `Fb i? i+ M #2 `2HB #Hv m
i?2 }2H/X h?2 #BHBiv iQ 2p Hm i2 HHQrb mb iQ +QKT `2 /Bz2`2Mi TT
r? i `2[mB`2b 7m`i?2` `2b2 `+? M/ /2p2HQTK2MiX h?2 +`2 iBQM Q
i i?2 v2 `Hv qQ`Fb?QT QM J +?BM2 h` MbH iBQM U F?# `/2? 2i HX
T`Q;`2bb QM i` MbH iBQM /B`2+iBQMb bm+? b 1M;HBb? iQ :2`K M M/ 1
HbQ b22BM; `2+2Mi rQ`F QM +`2 iBM; HQr@`2bQm`+2 i` MbH iBQM /
a Gh U F2` 2i HX- kykkc" #B`v2 2i HX- kykkV M/ i?2 K2`B+ bLGA U1
/ i b2ibX "2vQM/ i?2 }2H/ Q7 i` MbH iBQM- 2p Hm iBQM #2M+?K `Fb bm

b2`pB+2 T`QpB/2`b UGaSbVX h` MbH iQ`b `2 `2[mB`2/iQ #2 M iBp2 bT
i `;2i H M;m ;2 M/ ? p2 ?B;? H2p2H ~m2M+v BM 1M;HBb?X h` MbH i
i H2 bi irQ iQ i?`22 v2 `b Q7 i` MbH iBQM 2tT2`B2M+2 BM i?2 `2H2p
? p2 M + /2KB+ /2;`22 BM i` MbH iBQM Q` HBM;mBbiB+b M/ i?`22 iQ
2tT2`B2M+2 B7 i?2v /Q MQi? p2 Mv `2H2p Mi + /2KB+ [m HB}+ iBQMX
i` MbH iBQM i2bi 2p2`v R3 KQMi?b iQ bb2bb i?2B6 HiQ`M2b@kyy
`2pB2r2`b `2 HbQ `2[mB`2/iQ #2 M iBp2 bT2 F2`b Q7 i?2 i `;2i H M;m
? p2 i` MbH iBQM /2;`22- iH2 bi}p2 v2 `b Q7 2tT2`B2M+2 rQ`FBM; b
`2pB2r 2tT2`B2M+2- M/ r?2`2 TQbbB#H2 `2 ++`2/Bi2/#v `2H2p Mi
q2 MQi2 i? ii?2b2 `2 bi`BM;2Mi bi M/ `/bH
HQr@`2bQm`+2 H M;m ;2b BM i?2 7mim`2 K v #2 /B{+mHiX H`2 /v 7Q`
`2pB2r2`i? iK22ib i?2 +`Bi2`B #Qp2 Bb p2`v +? HH2M;BM;X AM i
i?2 [m HB}+ iBQM T`Q+2bb iQ ++2Ti TTHB+ iBQMb 7`QK `2pB2r2`b r
/2;`22b bm+? b GBM;mBbiB+b Q` 7`B+ M G M;m ;2 aim/B2b- Q` MQ /
? / 2ti2MbBp2 +QKK2`+B H i` MbH iBQM 2tT2`B2M+2 U2X;X =Ry v2
HQr@`2bQm`+2 H M;m ;2b BM i?2 7mim`2- r2 #2HB2p2 i? ii?2`2 `2 b
2tT2`B2M+2/ M/ bFBHH2/i` MbH iQ`b r?BH2 K BMi BMBM;?B;?[m H
QM2 Q7 bm+? bQHmiBQMb Bb iQ i` MbH i2 7`QK MQM@1M;HBb? bQm`+2
M/ /2b+`B#2 Bi BM ;`2 i2` /2i BH BM a2+iBQM 9XRXkX

6HQ`2b@kyy h` MbH iBQM h2 6HQ`2rX@kiyy+`2 iBQM rQ`F~Qr BM+Q`TQ
` i2b i?2 Q`BHBM2b@TRQR+2bb2b HQM;rBi? 72r M2r BMBiB H T? b2b
/2i BH #2HQrX

Ç HB;MK2Mi S?q2 2,p2 BMi`Q/m+2/ M BMBiB H HB;MK2Mi T? b2 iQ i?
i?2 i` MbH iQ`b M/ `2pB2r2`b #2HQ`2bi@kyy2H`2BM2,b2p2` H bi2Tb
BM+Q`TQ` i2/ BM HB;MK2Mi #2ir22M i?2 i` MbH iBQrM M/ [m HB
HB;MBM; QM `2bQm`+BM; M/ i `;2i `2;BQMb- HBM;mBbiB+ iQTB+b
`2pB2r2`b T2` H M;m ;2 i?`Qm;? M2r HB;MK2Mi i2KTH i2- M/ [m2
HBM;mBbib QM #Qi? bB/2bX h?2 HB;MK2Mi i2KTH i2 ?2HT2/HBM;m
i?2 H M;m ;2 b+`BTi- bi M/ `/Bx iBQM- bT2HHBM;- #Q``Qr2/i2`Kb-
+QMi2Mi bivH2- M/ `2bQm`+2b 7Q` ;HQbb `B2b- M/ b KTH2 +QMi2
h?Bb ? b #22M 2bT2+B HHv ?2HT7mH 7Q` H M;m ;2b rBi? H2bb
i` MbH iBQMX

Ç h` MbH iBQM Sh? b2,H iBQM i?2M #2;BMb rBi? M BMBiB H i` MbH iB
i?2 b K2 kyy b2Mi2M+2b `2 i` MbH i2/#v HH T `iB+BT iBM;i` MbH
h?2 BMBiB H i` MbH iBQM/ i +QMi BMb M 2p2M bTHBi +`Qbb i?2 i
qBFBDmMBQ`- M/ qBFBpQv ;2- rBi? i?2 b2;K2Mib +Q``2bTQM/BM;
+QMi2ti M/ +QMiBMmBivX h?2 BMBiB H i` MbH iBQMb `2 i?2M b
7Q` `2pB2rX h?2 K BM 7Q+mb Q7 i?2 BMBiB H i` MbH iBQM M/ Z
M/ HB;M QM i?2 i` MbH iBQM TT`Q +? #2ir22M i?2 i` MbH iQ`b
`2TQ`i +QMi BMb b2Mi2M+2@H2p2H 722/# +F UB/2MiB}2/ 2``Q` +
M/ +QKK2Mib r?2`2 TQbbB#H2V M/ ?B;?@H2p2H 722/# +F QM HQ
bT2+B}2/ b+`BTi- mb2 Q7 #Q``QrBM;b M/fQ` M2QHQ;BbKb- M K2/ 2
M/ `2;Bbi2`X

Ç Ai2` iBQM,MbH iBQM GaS i2 Kb K v `2bTQM/ iQ i?2 BMBiB H Z
`#Bi` iBQMX /DmbiK2Mib `2 i?2M K /2 iQ HH HB;MK2Mi K i2`B
i?2 i` MbH iBQM TT`Q +? Bb mT/ i2/ M/ `2@ HB;M2/ QMX h?2 7mH
b2Mi2M+2b i?2M #2;BMb Ub22 :Qv H 2i HX UkykkV 7Q` /2i BHbVX

Ç *QKTH2iBQM2, 7mHH i` MbH iBQM Bb +QKTH2i2/- i?2 Z GaS i2 K T
Z `2pB2r M/ bb2bb2b kyW b KTH2 / i X PTiBQM H `#Bi` iBQM-`
+?2+Fb K v 7QHHQr B7 i?2 }M H [m HBiv b+Q`2 Q7 i?2 i` MbH iBQM

9XRXk "2M+?K `F *`2 iBQM 7Q` LQM@1M;HBb? .B`2+iBQMb

h?2 bi M/6 /Q`2b@ Q~Qr 7Q+mb2b QM i` MbH iBQM QMHv 7`QK 1M;
bi M/ `/Bx2b i?2 T`Q+2bb 7Q` HH H M;m ;2b- Bi? b +H2 ` HBKBi iBQ
K Mv [m HB}2/ i` MbH iQ`b r?Q K v MQi bT2 F 1M;HBb?- #mi `2 #H2
b2p2` H MQM@1M;HBb? H M;m ;2bX 6m`i?2`- b2p2` H H M;m ;2b K v
MQM@1M;HBb? bQm`+2X AMbi2 /- r2 7Q+mb QM / Ti iBQM M/ i`
+mbiQKBx2/ Z rQ`F~Qrb iQ bmTTQ`i i?BbX

h` MbH iBQM Q7 ` #B+ GaS i2 Kb M Hvb2bBb rQ`F~Qr iQ +`2 i2/ i b2ib 7Q` p
p `B Mib Q7 ` #B+- 2tT M/BM; Qm` H M;m ;2 +Qp2` ;2 #2vQM/ JQ/2`
`2;BQM H p `B Mib bm+? b JQ`Q+ 6 MQ`2 #bQ7 2 +? ` #B+ H M;
i2 Kb M Hvx2/ i?2 HBM;mBbiB+ +? ` +i2`BbiB+b Q7 2 +? ` #B+ H M;
/Bz2`2/ 7`QK JQ/2`M ai M/ `/ ` #B+ QM p `BQmb HBM;mBbiB+ bT2-
/Bz2`2M+2b- ;` KK iB+ H M/ bi`m+im` H /Bz2`2M+2b- BM~m2M+2 7`Q
M/ BM7Q`K iBp2 +QMi2Mi bivH2X " b2/ QM i?2b2 M Hvb2b- ` #B+
i` MbH i2/ /B`2+iHv 7`QK 1M;HBb? Q` / Ti2/ 7`QK i?2 JQ/2`M ai M/ `/
i?2 1M;HBb? bQm`+2 T`QpBQ2/2b+ QHMM2miXQB/ i? iBKTH2K2Mi2/ / Ti
GaS i2 Kb HbQ +`2 i2/ i2`KHBbi +QMbBbiBM; Q7 i2`Kb 7`QK JQ/2`M a
2[mBp H2Mi i2`K BM i?2 i `;2i ` #B+ H M;mQB/iQ 2Mbm`2 +QMbBbi2M
hrQ iB2`b Q7 [m HBiv bb2bbK2Mi r2`2 +`2 i2/ 7Q` / Ti iBQM 7`Q
` #B+X PM2 iB2` 2M+QKT bb2/ T `iB H Z `2pB2r r?2`2 i?2 `2pB2r2` b
/ i M/ `2pB2r2/ i?2 i2`KHBbiX h?Bb T`Q+2bb r b TTHB2/ iQ H M;mQ
iQ ? p2 K BMHv pQ+ #mH `v /Bz2`2M+2b- bQK2 bi`m+im` H /Bz2`2M+2
Qi?2` `2;BQM H H M;m ;2bX MQi?2` iB2` `2[mB`2/ i?2 `2pB2r2` iQ QM
i?2 H M;mQB/b K BMHv /Bz2`2/ 7`QK JQ/2`M ai M/ `/ ` #B+ KBMBK H
mb ;2X h?2 NyW [m HBiv i?`2b?QH/ Bb TTHB2/ b mbm HX

a+`BTi h` MbHBi2h?2 Q2MrX`2 7Qm` H M2;m `;#bDMn -K#BMn -` #
i [nh7M;i? i r2`2 i` MbHBi2` i2/ 7`QK i?2B` G iBM b+`BTi +QmMi2`T `
GaS T2`7Q`Kb i` MbHBi2` iBQM BMiQ i?2 TT`QT`B i2 b+`BTibX h?2
b KTH2 Q7 i?2 i` MbHBi2` i2/ i2ti rBi? i?2 1M;HBb? bQm`+2 M/ G iBM b
+QMi2tiX AM i?2 Z `2TQ`i- i` MbHBi2` iBQM 2``Q`b `2 ~ ;;2/ QMHv
`2 MQ 2``Q` + i2;Q`B2b 7Q` i` MbHBi2` iBQM 2``Q`bX hrQ Q` KQ`2 2`
rQmH/ #2 ~ ;;2/ KBD?b2p2`Biv H2p2HX Mvi?BM; 72r2`KmBMQH/ #2 ~ ;;2
h?2 [m HBiv i?`2b?QH/ 7Q` i` MbHBi2` iBQM Bb N8WX

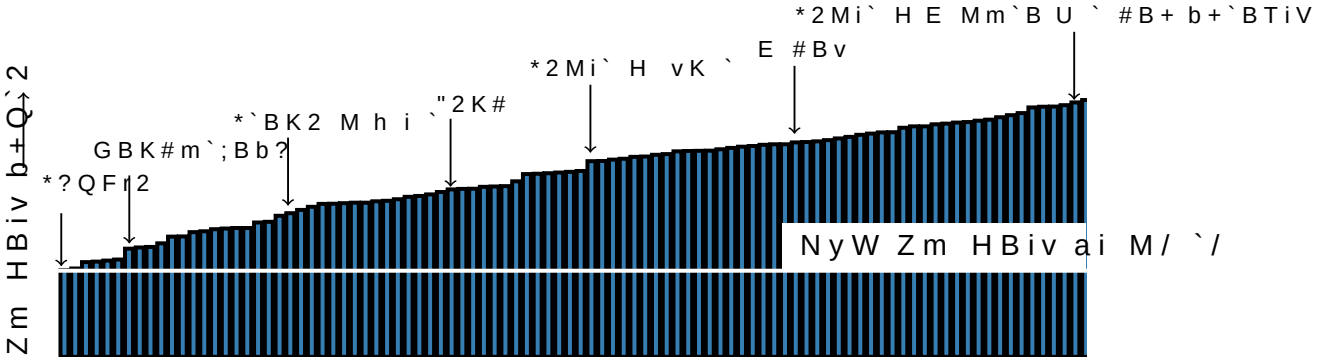
dX+Kn `-#+[n `-#2#n `-#M/`bn ` #2`2 / Ti2/X

Pp2`pB2r ai iBbiB+b	
O Q7 b2Mi2M+2b	
p; O Q7 rQ`/bfb2Mi2M+2	
O Q7 `iB+H2b	

aTHBi	O Q7 b2Mi2M+2b
/2p	NNd
/2pi2bi	RyRk
i2bi	NNk

O Q7 G M;m ;2b `2[mB`BM; _2@i` MbH iBQM	
jyyR	O Q7 _2@i` MbH iBQMb
J kR	O Q7 _2@i` MbH iBQMb
39k	
p; O Q7 . vb iQ h` MbH i2 R H M;m ;2	9k
p; O Q7 . vb iQ HB;M	k3
p; O Q7 . vb 7Q` R H M;m ;2	RRN
a?Q`i2bi hm`M `QmM/ U/ vbV 7Q` R H M;m ;2	
GQM;2bi hm`M `QmM/ U/ vbV 7Q` R H M;m ;2	

h #H2 6kGP_1a i :H M+2X 6GH2 71a Bb /BpB/2/ BMiQ i?`22 2p Hm iBQM
iQi HBM; jyyR b2Mi2M+2bX `v Q7 Zm HBiv *QMi`QH # b2/ QM i?2 bi iB
H M;m ;2b i? iBKTH2K 6H2/ 2b@kyy ~QrX



6B;m`2 Zm HBiv Q7 6GP_1a q@kyy B+i?2 [m HBiv bbm` M+2 b+Q`2 7Q`
H M;m ;2b HQM2b@kyy2 KBMBKmK ++2Ti #H2 bi M/ `/ Bb Ny T2`+2MiX

9XRXj 6HQ`2b@kyy i ;H M+2

Pp2`pB2r 6XHQ`2b@kyy MbBbib Q7 i` MbH iBQMb 7`QK 39k /BbiBM+ir2#
jyyR b2Mi2M+2bX h?2b2 b2Mi2M+2b `2 /BpB/2/ BMiQ i?`22 bTHBib,
`2H2 b2 i?2 7mHH i2ti Q7 i?2 /2p M/ /2pi2bi bTHBib- M/ F22T i?2 i2b
2p Hm iBQM b2`p2`X PM p2` ;2- b2Mi2M+2b `2 TT`QtBK i2Hv kR rQ
i?Bb BM7Q`K iBQM BM h #H2 k UH27iVX

Zm HBiv X +QMbB/2` H M;m ;2 `2 /v67HQ`2b@kyy MbBbib Q7 BM H ?mK M
[m HBiv bbm` M+2 2p Hm iBQMX q2 /BbTH v i?2 [m HBiv b+Q`2b Q7 Qm
b2p2` H 2t KTH2 H M;m ;2b H #2H2/X JBbi` MbH iBQM M/ mMM im` Hi
i?2 KQbi +QKKQM 2``Q`b 7QmM/ r?BH2 bb2bbBM; i?2 [m HBiv Q7 i?2 ?
r2`2 K BMHv /m2 iQ BM~m2M+2b 7`QK Qi?2` MQM@1M;HBb? H M;m ;2
mb2/ BM i?2 i `;2i +QKKmMBiB2b- H2 /BM; iQ 2t+2bbBp2 #Q``QrBM;b Q
HBi2` H i` MbH iBQMb /m2 iQ BM7`2[m2Mi mb ;2 Q7 i?2 i `;2i H M;m ;
+QMi2MibivH2 M/ i?2 HQr2` H2p2Hb Q7 bi M/ `/Bx iBQMX h?2`2? b HI
BM bT2HHBM; BM+QMbBbi2M+B2b BM i?2 ?mK M i` MbH iBQMb /m2 iQ
H2 /BM; iQ BM+QMbBbi2Mi Q` 2p2M bm#D2+iBp2 Q` T`272`2MiB H TT

*? HH2M;2b BM *`2 iBM; . i b2ib 7Q` o2`v GQr@_2bRQm2`+2HG M;m ;2
+QKT `2/6iHQQ`2b@RQmR` M2r i` MbH iBQM rQ`F~Qr bm#bi MiB HHv bi
i` MbH iBQM 2zQ`iX 6Q` 2t KTH2-i?2 MmK#2` Q7 H M;m ;2b `2[mB`BM;
`B;?iV Bb QMHv Ry-/Q6HQ``2QK@9RyQBM2p2`-/2bTBi2 i?2b2 BKT`Qp2K2M
+QMiBMm2/iQ 2tT2`B2M+2 bBK6BHQ``2+?@HmM2Mi;2b2 bM2 i2` b+ H2/m2
iQ i?2 BM+`2 bBM; HQr@`2bQm`+2 M im`2 Q7 i?2 H M;m ;2bX 6Q` 2t KT
`2 MQi b Q7i2M rQ`F2/rBi? BM i?2 HQ+ HBx iBQM Q` i` MbH iBQM
`2 HQr2` H2p2Hb Q7 BM/mbi`v@rB/2 bi M/ `/Bx iBQM-H2 /BM; iQ
M pB; i2 UaF /Bœ 2i HX-kyR9 VX h?Bb H2/iQ HQM;2`im`M `QmM/
}M/BM; M2r i` MbH iQ`b M/ `2pB2r2`b b2p2` H iBK2bX h?2b2 +? HH2
/m`BM; bQK2 Q7 i?2 KQ`2 /B{+mHi H M;m ;2b bm+? b aB+BHB M M/"m
bB;MB}+ MiHv HQM;2` T2`BQ/b Q7 iBK2 iQ +QKTH2i2 Uk3d / vbVX

9Xk LGG"a22/. i b2i

J +?BM2 H2 `MBM; Bb MQiQ`BQmbHv / i @?mM;`v-H2 /BM; iQ K Mv
`2/m+BM;i?2 KQmMi Q7 `2[mB`2/bmT2`pBbBQMX _2+2Mi /p M+2b B
2i HX-kykRc:m 2i HX-kyRNc CQ?MbQM 2i HX-kyRdc w? M;2i HX
H2 `MBM;U" TM 2i HX-kykkc GBm 2i HX-kykyc J 2i HX-kykRV-7Q
i?Bb `2HB M+2X >Qr2p2`-;2M2` iBQM i bFb bm+? b i` MbH iBQM HB
/2bB`2/[m HBiv H2p2Hb rBi?QmibQK2 bi `i2`/i X 6Q` BMbi M+2-Bi
;QQ/i` MbH iBQM rBi?Qmi b22BM; KBMBKmK MmK#2` Q7 b2Mi2M
aBKBH `Hv-Bi K v #2 /B{+mHi iQ +H bbB7v r?B+? H M;m ;2 b2Mi2I
`2HB #H2 2t KTH2b Q7 i2ti BM /Bz2`2Mi H M;m ;2bX @a22/i`;2i@bB/2/i BMp `BQmbH M;m ;2b+M #2miBHBx2/7Q`
Q7 T`Q72bbBQM HHv@i` MbH i2/b2Mi2M+2b BMjN H M;m ;2bX
am+? /i b2i? b MmK2`Qmb TQi2M LBG H"@a22/i`;2i@bB/2/i BMp `BQmbH M;m ;2b+M #2miBHBx2/7Q`
Bb /2}MBi2Hv BM i?2 bT2+B}2/H M;m ;2- b Bi Bb 7mHHv T`Q72bbB
LGG"@a22/i`;2i@bB/2/i BMp `BQmbH M;m ;2b+M #2miBHBx2/7Q`
KQ/2Hb i? i +H bbB7v r?B+? H M;m ;2 M `#Bi` `v TB2+2 Q7 BMTmi
+M HbQ #2mb2/7Q` Bib HB;M2/#Bi2ti-7Q` 2t KTH2 iQ i` BM i` MH
QTiBQM Bb iQGmi'B@HaBxQ//Q /QK BM }M2imMBM;-bm+? b /TiBM;;2M2`
i` MbH iBQM KQ/2Hb iQ i?2 qBFBT2/B /QK BMX

aQm`+2 a2Mi2M+2 a2.Hi2+7QLQGVX"@a22/i`;2i@bB/2/i BMp `BQmbH M;m ;2b+M #2miBHBx2/7Q`
Q7 `iB+H2b 2p2`v qBFBT2/B+QbH?QmHB?MpQ7 Ry-yyy qBFB/i A.b+Q``2
iQ MQi #H2 iQTB+b BM /Bz2`2Mi }2H/b Q7 FMQrH2/;2 M/?mK M +iB
RR+i2;Q`B2bSb2Q+PHB2biQS?BHQbQT?v M/2_Q2;HBTXBvQ2MmMB7Q`KHv
b KTH2/ bm#b2iQ7 A.b 7`QK r?B+? r2 rQmH//` r/i-M/K TT2/i?
`2bTQM/BM; 1M;HBb? qBFBT2/B `iB+H2bX 6`QK 2 +? Q7 i?2b2 `iB
/i i? irQmH/#2 b2Mi iQ i` MbH iQ`bX AMbi2 /Q7 2ti` +iBM; BM/Bp
rQmH/? p2H27i i` MbH iQ`b rBi? HBiiH2 +QMi2ti iQ rQ`F rBi?-r2 +?Q

3XLQi2 i? ir2 7Q+mb QMG1N@aQ2b/i?2b2 r2`2 i?2 H M;m ;2b r?2`2 i?2`2 /B/MQi2tBbi Tm
p BH #H2 ?B;?@[m HBiv #Bi2ti 7Q` i` BMBM; BM H `;2[m MiBiB2bX
NX?iiTb,ffK2i XrBFBK2/B XQ`;frBFBfGBbinQ7n `iB+H2bn2p2`vnqBFBT2/B nb?QmH/r

.QK BM a2H2+i2Q+MQXHH2+i2/ j-yyy 1M;HBb? b2Mi2M+2b BM 2 +? Q7 7Qm
M/ b2Mii?2Ki Q T`Q72bbBQM Hi` MbH iQ`bLiG#2@JMbHBti2/ BMiQ 2 +?
H M;m ;2bX h?2 i` MbH iBQM rQ`F~Qr mb2/ Bb M HGG"@aX2/i?2 QM2
h?2 /QK BMb BM+Hm/2/ `2,

- Ç L2rb, q2 i` MbH i2 i?2 1M;HBb? bB/2 Q7 i?2 qJhkR 1M;HBb?@:2`K
  b2i- +QMi BMBM; b KTH2 Q7 M2rbT T2`b 7`QK kyky U F?# `/2? 2i
- Ç a+`BTi2/ 7Q`K H ,bTq2 i2 ?MbH i2 i2ti 2ti` +i2/ 7`QK b2`B2b Q7 b
  1M;HBb?@H M;m ;2 i HFb +Qp2`BM; p `B2iv Q7 iQTB+bX
- Ç lMb+`BTi2/ BM7Q`K ,Hqb2T2i2`+?i j-yyy mii2` M+2b 7`QK i?2 KmHiB@
  +? i/ i b2iQ7 sm 2i HX UkykkV- r?B+? +QMi BMb QM p2` ;2 kj rQ
- Ç >2 Hi,?q2 i` MbH i2/ QM2 qQ`H/ >2 Hi? P`; MBb iBQM `2TQ`i U.QM H
  kyRdV M/ +QK#BM2/ BirBi? b2Mi2M+2b i` MbH i2/ 7`QK i?2 1M;HBb
  *Q`QM *`BbBb Rk_2TQ`iX

9X9 *QM+HmbBQM

hQ bmKK `BHkQ-2b@kyy+? 2M #H2b `2HB #H2 2p Hm iBQM Q7 Qp2` k
+`BiB+ H 7Q` 2Mbm`BM; i?2 [m HBiv Q7 i?2 L2bGn"@a2Hm vbvbM2Kb ;2M
BKTQ`i Mi `QH2 7Q`i` BMBM; #Qi? b2Mi2M+2 2M+Q/2`b Ub22 a2+iBQ
Ub22 a2+iBQM eX8VX 6BMGH"@vJ.Q2Km2iBmBx2?2 ;2M2` HBx #BHBiv Q
i` MbH iBQM KQ/2Hb +`Qbb KmHiBTH2 /QK BMb Ub22 a2+iBQM 3XjV
i?2 +`2 iBQM Q7 i?`22 ?mK M@i` MbH i2/ / i b2ib M/ i?2B` mb2b- r2
i` BMBM; / i 7Q` Qm` 2zQ`i BM i?2 bm#b2[m2Mi b2+iBQMX

8X miQK iB+ HHv *`2 iBM; h` MbH iBQM h` BMBM; . i 7Q` >mM G M;m ;2b

h?2 +m``2Mi i2+?MB[m2b mb2/ 7Q` i` BMBM; i` MbH iBQM KQ/2Hb `2
`2bQm`+2 b2iiBM;b i? iBb- r?2M / i 7Q` H M;m ;2 Bb HBKBi2/ BM
/ i Bi2tQ` T B`b Q7 i` MbH i2/ b2Mi2M+2bV MQMBQMHBQ;HM;m ;2 / i U
BM QM2 H M;m ;2 QMHvVX AM 7 +i- K Mv HQr@`2bQm`+2 H M;m ;2b
bK HH i `;2i2/ #Bi2ti / i b2ib bm+? b i?2 *?`BbiB M "B#H2 UJ+* `i?v
`2 2ti`2K2Hv HBKBi2/ BM /QK BM /Bp2`bBivX AM i?Bb b2+iBQM- r2 /
b+ H2 / i b2ii? i +Qp2`b ?mM/`2/b Q7 H M;m ;2b M/ /Bb+mbb i?2 +?
MQBbv / i ir2#@b+ H2X

6Q` +QMi2ti- Tm#HB+Hv p BH #H2 #Bi2ti / i Bb Q7i2M b+ `+2 U:C
TT`Q +? +2Mi2`b `QmM/ 2ti2M/BM; 2tBbiBM; / i b2ib #v +QHH2+iBM;
/ i M/ mbBM; H `;2@b+ H2 / i KBMBM; Ua+?r2MF 2i HX- kykR#V iQ B/2
?B;? T`Q# #BHBiv Q7 #2BM; i` MbH iBQMb Q7 2 +? Qi?2` BM /Bz2`2M
7Q` ?mM/`2/b Q7 H M;m ;2b- r2 }`bi /2p2HQT H M;m ;2 B/2MiB}+ iBQM b
i? i H #2H r?B+? H M;m ;2 ;Bp2M TB2+2 Q7 i2ti Bb r`Bii2M BMX a
p BH #H2 KQMQHBM;m H / i - TTHv b2Mi2M+2 bTHBiiBM; M/ GA.

Rk XiiTb,ffK/Xi mbXM2if+Q`QM

6B;m`2mi QK iB+ . i b2i *`2 iBQM *QMi`B#miBQMb Q7 LQ G M;m ;2 G2
"2?BM/ b ?B;?HB;?i2/- r2 +`2 i2 H M;m ;2 B/2MiB}+ iBQM M/ KQMQHE
T`Q+2bb- i?2M /2b+`B#2 i?2 i` BMBM; Q7 G a1_j iQ T`Q/m+2 H `;2@b-
?mM/`2/b Q7 H M;m ;2bX

K2+? MBbKb Ua2+iBQM 8XkV- M/ i?2M KQp2 ?2 / rBi? KBMBM; HB;M2
Qp2`pB2r Q7 i?Bb T`Q+2bb Bb T`2b2Mi2/ BM 6B;m`2 dX

8XR G M;m ;2 A/2MiB}+ iBQM

G M;m ;2 B/2MiB}+ iBQM UGA.V Bb i?2 i bF Q7 T`2/B+iBM; i?2 T`BK
Q7 i2tibX Ai Bb rB/2Hv mb2/ BM +QKK2`+B /2i2T+TiHHBM7mBQMb2Ubm+? b
2K#2//2/ BM bQK2 r2# #`Qrb2`bV M/ Bb Q7 T `iB+mH ` BKTQ`i M+2
T`Q+2bbBM; `2b2 `+?X h?2 `Bb2 Q7 H `;2@b+ H2 T`2i` BMBM;- T `iB+
KmHiBHBM;m H KQ/2Hb- Bb bi`QM;Hv /2T2M/2Mi QM i?2 2tBbi2M+2 M/
/ i i b+ H2X /p M+2b BM +`Qbb@HBM;m H `2T`2b2Mi iBQM H2 `M
kyRNc q M; 2i HX- kyky#V bm+? b H `;2@b+ H2 #Bi2ti KBMBM; U" ¢
2i HX- kykkc a+?r2MF 2i HX- kykR#V- mMbmT2`pBb2/ K +?BM2 i` MbH
_2M 2i HX- kyRNc u M; 2i HX- kyR3V M/ # +F@i` MbH iBQM i b+ I
`2[mB`2 H `;2 [m MiBiB2b Q7 +H2 M KQMQHBM;m H / i X h?2b2 /BbT
Qm` 7Q+mb QM H `;2@b+ H2 / i KBMBM; Q7 HB;M2/ b2Mi2M+2b- BMp
BMTmi i2ti Q7i2M /` rM 7`QK r2# +Q`TQ` bmM? Hb#2QHBKQ;M?2kH Bi?
+Q``2bTQM/BM; H M;m ;2bX

h?2`2 `2 72r r2HH@FMQrM +? HH2M;2b bbQ+B i2/ rBi? H `;2@b+
B/2MiB}+ iBQM mbBM; r2# / i U* bF2VQk 2BiMHXB-bkyk+y ,+QmH/ Q++m` /m
iQ i?2 b+ `+Biv Q7 i2ti `2HB #Hv H #2H2/ #v H M;m ;2X 6Q` 2t KTH2-

RjX?iiTb,ff+QKKQM+` rHXQ`;f

6B;m`2Pd2`pB2r Q7 Qm` "Bi2ti JBMBMG; MB;m2;H2BM2XiB}+ iBQM Bb TTH
QM r2# +Q`TQ` iQ 2ti` +i KQMQHBM;m H b2Mi2M+2bX HB;M2/ T B`b
G a1_jX

#22M i` MbH i2/ BMiQ rB/2 `` v Q7 H M;m ;2bX >Qr2p2`- M GA. bv
+Q`Tmb rQmH/ MQi`2HB #Hv +H bbB7v b2Mi2M+2b 7`QK MQM@"B#HB
i` BMBM;/ i Bb MQii`BpB H, r?BH2 i?2 r2# +QMi BMb/ i BM i?Qmb M
2i HX- kyR3c a+ MM2HH- kyydV- KQbi Q7 Bi Bb mMH #2H2/X 6BHHBM;
Bb 7`2[m2MiHv mb2/ 7Q` i` BMBM; H M;m ;2 B/2MiB}+ iBQM Uh?QK
#2vQM/i?2 *?`BbiB M "B#H2 U Hi?Qm;? bm+? `2H iBp2Hv +H2 M 7Q`K
Q7 i?2 r2# i HUk2Vpc2`2 +H bb BK# H M+2 +QmH/2tBbi#2+ mb2 K Mv Q7
H M;m ;2b Q7 BMi2`2bi iQ mb ? p2 HQr T`2b2M+2 QM i?2 r2#X 6Q` +H b
? p2 M 2ti`2K2Hv HQr 7 Hb2 TQbBiBp2 ` i2X Pi?2`rBb2- HQr@`2bQm
KBbB/2MiB}U+jVLE{QBM2cM+v iQ `mM Qp2` H `;2 r2# +QHH2+iBQMb `2K BMb
+H bbB}+ iBQM Bb K bbBp2Hv T ` HH2HBx #H2-`mMMBM;Bi QM HH i2
b2+iBQM-r2/2b+`B#2 Qm` TT`Q +?iQ H M;m ;2 B/2MiB}+ iBQM M/?
H M+2 #2ir22M T`2/B+iBp2 T2`7Q`K M+2 M/b+ H #BHBivX

8XRXR _2H i2/qQ`F

h?2`2 Bb 2ti2MbBp2 HBi2` im`2 /2/B+ i2/iQ i?2 /2p2HQTK2Mi Q7 GA. bv
UkyRNV ;Bp2 `2+2Mi M/ +QKT`2?2MbBp2 Qp2`pB2r Q7 i?2 72 im`2
i?2 HBi2` im`2X q?BH2 GA. +QmH/#2 b22M b bQHp2/T`Q#H2K BM
kyy8V- Bi `2K BMb M QT2M +? HH2M;2 7Q` r2#/i U #/DB 2i HX-
kykyc w KTB2`B 2i HX- kyR8#VX aT2+B}+ UHHBv+ BBbM;2bm+Q2Hb27+2H
TT`Q +?2biQ KQ`2H M;m ;2bUC mU?kBMB+M2/M2MibH?X2`kyiR?d2V2 Bb bB;MB}+
/QK BM KBbK i+? UqB//Qrb M/"`2r- kykRV BM i?2 + b2b Q7 b?Q`i i
H M;m ;2b U.mp2M? ;2UjkyBRNBMd;mMBb?BM;bBKBH `H M;m ;2b U:Qmii2

a+ HBM;GA. iQ JQ`2 G M;2mQ2bX ii2MiBQM iQ /p M+2 GA. i2+?MB[m2
H2/iQ MQiB+2 #H2 BM+`2 b2 BM #Qi? H M;m ;2 +Qp2`R9;2 M/ ++m
M/7 bii2tU:` p2 2i HX- kyR3V `2 irQ`2 /BHv p BH #H2 KQ/2Hb Qz2`
T2`7Q`K M+2 7Q`Ryd M/R3dH M;m ;2b `2bT2+iBp2HvX "v mbBM;MmK2

R9XiiTb,ff;Bi?m#X+QKf;QQ;H2f+H/j

UkykyV M/"`QrM UkyR9V `2TQ`i2p2M ?B;?2` +Qp2` ;2-bmTTQ`iBM;
`2bT2+iBp2HvX h? ib B/-/2p2HQTK2Mib `QmM/HQr@`2bQm`+2H M;m
/m2iQi?2 2KT? bBb QM`2HB;BQmbi2tib M/+QMbi` BMib#`Qm;?i #Qm
* br2HH 2i HX UkykyV b+ H2 mTiQ R-ekN H M;m ;2b mbBM; rQ`/HBl
#QQibi` Ti` BMBM;/i 7QmM/QMi?2r2#X h?2b2 TT`Q +?2bmbBM;
/QK BM BK# H M+2, #2+ mb2i?2 p BH #H2i2ti/QK BMb p `v #v H
+QM~ i2b/QK BM rBi? H M;m ;2X AM H+Q M2b@bQ ymb2+mb`/2p2HQTK2Mi
b2i-bQi? iQm` GA. bvbi2K T2`7Q`K M+2 BbimM2/Qp2` mMB7Q`K/Q
+Qm`b2mb2i?2 *?`BbiB M "B#H2 b mMB7Q`K6/HQ[ BM@BkyQr2p2`-r2
+HQb2`iQ r2#+QMi2MiX

.QK BM JBbK i+"2X+ mb2i?2 r2#+Qp2`b p2`v#`Q/b2iQ7/QK BMb
H #2H2/i2ti Bb b+ `+2-i?2`2 Bb HKQbi Hr vb /QK BM KBbK i+?#2ir2
i?2 r2#i2ti#2BM;+H bbB}2/X qB//Qrb M/"`2r UkykRV/2b+`B#2 M2r7
` MF Q7 rQ`/b rBi?BM 7`2[m2M+v i #H2b i? i2M? M+2b `Q#mbiM2bb Q
KBbK i+?2bX h?2v i` BM i?2B` +H bbB}2` QM qBFBT2/B M/`2TQ`i`
b2i-mM7Q`imM i2Hv+Qp2`BM; QMHv kk H M;m ;2bX a?Q`ii2ti Bb i+
7Q` aQmi? 7`B+ M G M;m ;2b rBi? bi +F2/+H bbB}2`X L2m` H M2irQ
HbQ/2`Bp2/BM Mb `B 2i HX UkykRVc a?2F? ` 2i HX UkykyV iQ ? M/H
1M;HBb? M/AM/B M H M;m ;2b U+Q/2 KBtBM;VX * br2HH 2i HX Ukyk
+H bbB7v 7 BHm`2 KQ/2b Q7 H M;m ;2 B/2MiB}+ iBQM QM r2#+Q`TQ` X
Q7 }Hi2`b HQM;rBi? M2r mMbmT2`pBb2/H2 `MBM; TT`Q +?iQ/` bi
iHBKBi2/+Qbi QM`2+ HHX h?2b2 }Hi2`b `2+QbiHv iQ/2pBb2 M/imM
aQK2Q7 i?2K r2`2 bm++2bb7mHHv TmiBMiQ T` +iB+2BM # /DB2i HX
p2`bBQM Q7 i?2 Pa* _ / i b2iX Pm` TT`Q +? +7QbK#2tM2`bp2/ i @/`Bp2
2i HX-kyR3V KQ/2Hb iQ ? M/r`Bii2M `mH2biQ bK HH b2iQ7 ? M/r`Bii2M `mH2biQ
?mK M 722/# +F QM +H bbB}+ iBQM 2``Q`bX

> M/HBM; aBKBH ` G M;m ;2b rBi?BM;#2ir22M bBKBH ` H M;m ;2b
+iBp2`2b2 `+?iQTB+-7Q` BMbi M+2-rBi?i?2b? `2/ibF QM .Bb+`BI
G M;m ;2b rBi?BM i?2 o `.B H rQ`Fb?QT U:Qmii2 2i HX-kyReVX a2p
H2 `MBM; H;Q`Bi?Kb HQM;rBi? bi M/`/M2m` H M2irQ`Fb `2+QKT
+xvMbFB UkykRV 7Q` LQ`/B+H M;m ;2bX .mp2M? ;2 UkyRNVc :Qmii2 2
UkyR8V 2tTHQ`2p `BQmb?B2` `+?B+H TT`Q +?2bi? i}`biT`2/B+ii?
i2ti-i?2M TTHv KQ`2bT2+B HBx2/+H bbB}2` iQ /BbiBM;mBb? #2ir2
;`QmTX AM i?Bb rQ`F-r2+QHH #Q` i2BM +HQb2T `iM2`b?BT rBi? HB
H M;m ;2b + M #2 2 bBHv+QM7mb2/ M/ M Hvx2i?2KQ/2H T2`7Q`K M
+H bbB}+ iBQM bi` i2;vX

8XRXk JQ/2Hb

q2 miBH7Bbxi2tiQ i` BM H M;m ;2B/2MiB}+ iBQM KQ/2Hb U"QD MQrbFB 2
2i HX-ky7RdV2tBb rB/2Hv mb2/7Q` i2ti+H bbB}+ iBQM i bFb/m2iQ B
bT22/-r?BH2 +?B2pBM;;QQ/[m HBivX q2 2K#2/+? ` +i2`@H2p2H M@
i?2M H2p2` ;2 KmHiB@+H bb HBM2 ` +H bbB}2` Q7MbiQ2TX h?2 HB;?
2M #H2b Qm` GA. KQ/2Hb iQ ? M/H2 r2#@b+H2/i X //BiBQM HHv
#2M2}iQ7#2BM;2 bBHv2tTH BM #H2-HHQrBM;mb iQ i` +2 Mv+H

`QQi + mb2X h?Bb Bb BMbi`mK2Mi H BM //`2bbBM; +QKKQM TBi7 H
H M;m ;2 QM r2# +Q`TQ` U* br2HH 2i HX- kykyVX

*H bbB}2` .2bB2MX T2`BK2Mi2/ rBi? irQ /Bz2U2M+iQ2K#BM;MbX Q7
KmHiBTH2 #BM `v +H bbB}2`b r?2`2 i?2 }M H /2+BbBQM Bb Q#i BM2
? pBM; i?2 ?B;?2bi b+Q`2 7i2` i?`2b?QH/ Bb TTHB2/X q2 TTHv i?`
i? i r?2M i?2 +QM}/2M+2 Q7 +H bbB}2` Bb HQr- i?2 +Q``2bTQM/BM; H
7Q` i?2 }M H /2+BbBQMX A7 MQM2 Q7 i?2 +H bbB}2`b bm`T bb Bib i?`
QmUXkVKmHiB+H bb +H bbB}2` mbBM; bQ7iK t Qp2` HH TQbbB#H2 H
i?`2b?QH/ QTiBKBx iBQM Bb /QM2 7i2` i?2 bQ7iK tX

Pm` 2tT2`BK2Mib KQiBp i2/ mb iQ 7Q+mb QM i?2 b2+QM/ TT`Q +?
/p Mi ;2bX 6B`bi- +? M;BM; i?2 i?`2b?QH/ 7Q` QM2 H M;m ;2 /Q2b MQi
Q7 i?2 Qi?2` - r?BH2 i?Bb Bb MQi i`m2 BM i?2 }`bi b2iiBM;X a2+QM/- r2
;2M2` HBx2b #2ii2` iQ Qmi Q7 /QK BM/ i r?B+? Bb Q!m`q2#BK `v mb2 +
/ i VX 6BM HHv- bBM;H2 +H bbB}2` ? b i?2 //2/ #2M2}i Q7 #2BM; +
i?mb bi`2 KHBMBM; i?2 H M;m ;2 B/2MiB}+ iBQM T`Q+2bbX

h` BMBM; . i M/ > M/HBM; J bbBp2 *H bbq2Ak#b2HTM#2X8+Hv p BH #H
/ i b2ib iQ i` BM Qm` GA. bvbi2K- T `iB HHv +Qp2`BM; Qm` Q7 BMi2`2
rBi?1 GG"@a2U2b/22 a2+iBQM 9XkV 7Q` Mv KBbbBM; H M;m ;2X >Qr2p
/ i p BH #H2 7Q` 2 +? H M;m ;2 Bb 7 `7`QK mMB7Q`K- M/ K bbBp2
` r i` BMBM; / i 2tBbib U* br2HH 2i HX- kykyc .mMM- kykyVX 6Q` 2
`2T`2b2Mib RyXRW Q7 Qm` i` BMBM; / i -r?BH2 JBM M;F # mUG iBM b+
6QHHQrBM; `Bp x? ; M 2i HX UkyRNV- r2 2tT2`BK2Mi2/ rBi? KmHiBTH
mTb KTHBM; 7Q` mM/2` `2T`2b2Mi2/ -r?2`2lb2M2i2M+2b 7i2` mTb KTHBM; M/ TB+F2/ K
Q7 i?2 / i b2i `2 b KTH2/ T`Q/ TXQ`PBQMK HHv 2iQ7Q`K M+2 r b Q#i BM2
$\frac{1}{T} = 0:3X$

h` BMBM; S ` K2iP2mbX#2bi KQ/2H r b i` BM2/ rBi? bQ7iK t HQbb Qp2`
rBi? H2 `MBM; ` i2 Q7 yX3 M/ 2K#2//BM; b rBi? k8e /BK2MbBQMbX q2
H2bb i? M i?Qmb M/ Q++m``2M+2b 7i2` mTb KTHBM; M/ TB+F2/ K
+? ` +i2` M@;` K H2M;i? Q7 irQ M/ }p2 `2bT2+iBp2Hv- r?B+? r2`2 bbB
Q7 bBx2 R-yyy-yyyX HH ?vT2`T ` K2iQ``2bbr@2Xym M2/ QM

8XRXj AKT`QpBM; GA. rBi? GBM;mBbiB+ M HvbBb

G M;m ;2 B/2MiB}+ iBQM Bb +? HH2M;BM; i bF r?2`2 MmK2`Qmb 7
2t +2`# i2/ #v i?2 ; T #2ir22M i?2 +H2 M / i i? iGA. KQ/2Hb `2i` BM2/
/ i i? iGA. KQ/2Hb `2 TTHB2/ iQX GA. KQ/2Hb i? i `2i` BM2/ BM b
QM ~m2MiHv r`Bii2M b2Mi2M+2b K v ? p2 /B{+mHiv B/2MiB7vBM; ;` K
BM+QKTH2i2 bi`BM;b 2ti` +i2/ 7`QK i?2 r2#X 6m`i?2`KQ`2- KQ/2Hb +
+Q``2H iBQMb i? i `2 MQi K2 MBM;7mH 7Q` i?2 i bF Bib2H7X AM H
+QHH #Q` i2/ +HQb2Hv rBi? i2 K Q7 HBM;mBbib i?`Qm;?Qmi /Bz2`2Mi
iQ B/2MiB7v T`QT2` `2 b Q7 7Q+mb- KBiB; i2 Bbbm2b- M/ 2tTHQ`2 b

GA. AMbT2+iBQM AM2`H2p2`X;2/ i?2 HB7Mi2iBtiQ#mBH/ M 2 bv@
iQ@mb2 BMi2`7 +2 7Q` HBM;mBbib iQ T22F BMiQ Bib BMM2` rQ`FBM;

6B;m`2G3A,. AMbT2+iBQM AmMi2 7Q+M2-M 2t KTH2 b2Mi2M+2 7`QK i?2 1M;
qBFBT2/B +QMi BMBM; b?Q`iT bb ;2 BM 6`2M+?X h?2 iQT k H #2Hb
`2 /BbTH v2/- HQM;rBi?i?2B` b+Q`2X L@;` Kb i? i+QMi`B#mi2/ i?2
Q` M2; iBp2HvV iQ i?2 T`2/B+iBQMb `2 ?B;?HB;?i2/ UBM;`22M M/ `2

M Hvx2 KQ/2H 2``Q`b M/ /Bb+2`M KQ/2H T ii2`MbX b BHHmbi` i2/
?Qr Km+? 2 +? M@;` K +QMi`B#mi2/ iQ i?2 }M H T`2/B+iBQMX AM QM
iQQH H2/ HBM;mBbib iQ MQiB+2 i?2 bBKBH `Biv BM T?QMQi +iB+b
AM/QM2bB M-r?B+? `2 QM2 Q7 i?2 KQbi7`2[m2MiHv +QM7mb2/H M
i?`Qm;? HBM;mBbiB+ `2b2 `+?i? iBM bTBi2 Q7 Q#pBQmb /Bz2`2M+2
BMi2HHB;B#BHBiv 2tBbib #2ir22M i?2 irQX

6BHi2`BM; h` BMBM;Q.KiBXB; i2 i?2 H2 `MBM; Q7 bTm`BQmb +Q``2H iB
i` BMBM; b KTH2b r?BH2 KQ/2HBM; ?mM/`2/b Q7 H M;m ;2b-r2 rQ`F
HBM;mBbib iQ /2p2HQT b2p2` H }Hi2`b-BHHmbi` i2/ BM h #H2 j M/
bm#b2[m2MiHv TTHB2/ QM Qm`` ri` BMBM; / i b2iX

Ç *? ` +i2` .Bbi`B#miBQM,6H?2i2`mB#H;B+ / i b2ib r2 mb2/ 7Q` i` BMBM
KQbiHv #mBHi 7`QK r2#T ;2bX h?`Qm;? BMp2biB; iBQM #v HBM;mBl
Q7 KBbH #2H2/ b2Mi2M+2b r2`2 7QmM/-HBF2Hv + mb2/ #v b?Q`
H M;m ;2rBi?BM T ;2-bm+? b AM/QM2bB M bBi2bi? i/BbTH v
TQ2KbX q2 HbQ MQiB+2/` M/QK +`2 iBp2 mb2 Q7 mM2tT2+i2/ b+`
/2+Q` iBQM Q` 2KT? bBb b TQBMi2/ QmiBM * br2HH 2i HX Ukyk
2t KTH2bX hQ //`2bb i?Bb T`Q#H2K-r2 b2 `+?2/ 7Q` /Bbi`B#miBQ
2Bi?2` #v +QKTmiBM; +? ` +i2` ?BbiQ;` Kb Q` #v HQQFBM; i i?2
b+`BTi mMB+Q/2 ` M;2X

*? ` +i2` >BbiQ;; kq2 +QKTmi2/ i?2 +? ` +i2` /Bbi`B#miBQMb
H M;m ;2 QM Qm` /2p2HQTK2Mi b2i M/+22}Ti2/ +M? ``#BQ™ bv2i
2 +? Q7 i?2K #v +QMbB/2`BM; HH +? ` +i2`i? 7 2H+H2BMMBHB2iX?BqM i?2
+QMb2[m2MiHv }Hi2`2/ Qmi Mv b2Mi2M+2 7`QK Qm` i` BMBM; b2i i?
i? M 3yW Q7 bm+? ++2Ti2/ +? ` +i2`bX

a+`BTi .2i2+,iBQMH M;m ;2b r?Qb2 b+`BTi bT Mb i?Qmb M/b Q7 +
+? ` +i2` ?BbiQ;` K K2i?Q/K2MiBQM2/ #Qp2 r b MQi b 2z2+iBp2 b
/Bbi`B#miBQM i`2M/b r2`2 H2bb T`QKBM2MiX b M Hi2`M iBp2-HB
` M;2b iQ /2}M2 ++2Ti2/ +? ` +i2` b2ibX Mv b2Mi2M+2 +QMi BM
+? ` +i2`b 7`QK i? i b2i r b 2p2Mim HHv /Bb+ `/2/X 6Q` 2t KTH2-i
BM h #H2 j 7Q` C T M2b2 M/ *?BM2b2 /QC TMQM/>QMbi+BBMTii?2X`B;?i